

Temporal transcriptomic dynamics in developing macaque neocortex

Reviewed Preprint

Revised by authors after peer review.

About eLife's process

Reviewed preprint version 2
February 13, 2024 (this version)

Reviewed preprint version 1
September 25, 2023

Posted to preprint server
July 14, 2023

Sent for peer review
July 12, 2023

Longjiang Xu, Zan Yuan, Jiafeng Zhou, Yuan Zhao, Wei Liu, Shuaiyao Lu, Zhanlong He, Boqin Qiang, Pengcheng Shu ✉, Yang Chen ✉, Xiaozhong Peng ✉

Institute of Medical Biology Chinese Academy of Medical Sciences, Chinese Academy of Medical Science and Peking Union Medical College, Kunming, China • Department of Molecular Biology and Biochemistry, Institute of Basic Medical Sciences, Medical Primate Research Center, Neuroscience Center, Chinese Academy of Medical Sciences, School of Basic Medicine Peking Union Medical College, Beijing, China • Agricultural Bioinformatics Key Laboratory of Hubei Province, Hubei Engineering Technology Research Center of Agricultural Big Data, College of Informatics, Huazhong Agricultural University, Wuhan, China • State Key Laboratory of Common Mechanism Research for Major Diseases, Chinese Academy of Medical Sciences, Peking Union Medical College, Beijing, China • Chinese Institute for Brain Research, Beijing, China • State Key Laboratory of Respiratory Health and Multimorbidity, Chinese Academy of Medical Sciences, Peking Union Medical College, Beijing, China • Institute of Laboratory Animal Science, Chinese Academy of Medical Sciences & Peking Union Medical College, Beijing, China

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

Abstract

Despite intense research on mice, the transcriptional regulation of neocortical neurogenesis remains limited in humans and non-human primates. Cortical development in rhesus macaque is known to recapitulate multiple facets of cortical development in humans, including the complex composition of neural stem cells and the thicker supragranular layer. To characterize temporal shifts in transcriptomic programming responsible for differentiation from stem cells to neurons, we sampled parietal lobes of rhesus macaque at E40, E50, E70, E80, and E90, spanning the full period of prenatal neurogenesis. Single-cell RNA sequencing produced a transcriptomic atlas of the developing rhesus macaque neocortex parietal lobe. Identification of distinct cell types and neural stem cells emerging in different developmental stages revealed a terminally bifurcating trajectory from stem cells to neurons. Notably, deep-layer neurons appear in the early stages of neurogenesis, while upper-layer neurons appear later. While these different lineages show overlap in their differentiation program, cell fates are determined post-mitotically. Pseudotime trajectories from vRGs to oRGs revealed differences in dynamic gene expression profiles and identified divergence in their activation of BMP, FGF, and WNT signaling pathways. These results provide a comprehensive picture of the temporal patterns of gene expression leading to different fates of radial glial progenitors during neocortex layer formation.

eLife assessment

This study presents a **useful** resource for the gene expression profiles of different cell types in the parietal lobe of the cerebral cortex of prenatal macaques. The evidence supporting the claims of the authors is **solid**, and revision has clarified some of the cell isolation and cell classification issues flagged by reviewers. This dataset will be of interest to developmental neurobiologists and could potentially be used for future comparative studies on early brain development.

Introduction

The neocortex is the center for higher brain functions, such as perception and decision-making. Therefore, the dissection of its developmental processes can be informative of the mechanisms responsible for these functions. Several studies have advanced our understanding of the neocortical development principles in different species, especially in mice. Generally, the dorsal neocortex can be anatomically divided into six layers of cells occupied by distinct neuronal cell types. The deep-layer neurons project to the thalamus (layer VI neurons) and subcortical areas (layer V neurons), while neurons occupying more superficial layers (upper-layer neurons) preferentially form intracortical projections¹. The generation of distinct excitatory neuron cell types follows a temporal pattern in which early-born neurons migrate to deep layers (i.e., layers V and VI), while the later-born neurons migrate and surpass early-born neurons to occupy the upper layers (layers II-IV)². In *Drosophila*, several transcription factors are sequentially explicitly expressed in neural stem cells to control the specification of daughter neuron fates, while very few such transcription factors have been identified in mammals thus far. Using single-cell RNA sequencing (scRNA-seq), Telley and colleagues found that daughter neurons exhibit the same transcriptional profiles of their respective progenitor radial glia, although these apparently heritable expression patterns fade as neurons mature³. However, the temporal expression profiles of neural stem cells and the contribution of these specific temporal expression patterns in determining neuronal fate have yet to be wholly clarified in humans and non-human primates. Over the years, non-human primates (NHP) have been widely used in neuroscience research as mesoscale models of the human brain. Therefore, exploring the similarities and differences between NHP and human cortical neurogenesis could provide valuable insight into unique features during human neocortex development.

In mammals, radial glial cells are found in the ventricular zone (VZ), where they undergo proliferation and differentiation. The neocortex of primates exhibits an extra neurogenesis zone known as the outer subventricular zone (OSVZ), which is not present in rodents. As a result of evolution, the diversity of higher mammal cortical radial glia populations increases. Although ventricular radial glia (vRG) is also found in humans and non-human primates, the vast majority of radial glia in these higher species occupy the outer subventricular zone (OSVZ) and are therefore termed outer radial glia (oRG). Outer radial glial (oRG) cells retain basal processes but lack apical junctions⁴ and divide in a process known as mitotic somal translocation, which differs from vRG⁵.

VRG and oRG are both accompanied by the expression of stem cell markers such as *PAX6* and exhibit extensive self-renewal and proliferative capacities⁶. However, despite functional similarities, they have distinct molecular phenotypes. Previous scRNA-seq analyses have identified several molecular markers, including *HOPX* for oRGs, *CRYAB*, and *FBXO32* for vRGs⁷. Furthermore, oRGs are derived from vRGs, and vRGs exhibit obvious differences in numerous cell-extrinsic mechanisms, including activation of the *FGF-MAPK* cascade, *SHH*, *PTEN/AKT*, and *PDGF*

pathways, and oxygen (O₂) levels. These pathways and factors involve three broad cellular processes: vRG maintenance, spindle orientation, and cell adhesion/extracellular matrix production⁸. Some transcription factors have been shown to participate in vRG generation, such as INSM and TRNP1. Moreover, the cell-intrinsic patterns of transcriptional regulation responsible for generating oRGs have not been characterized.

ScRNA-seq is a powerful tool for investigating developmental trajectories, defining cellular heterogeneity, and identifying novel cell subgroups⁹. Several groups have sampled prenatal mouse neocortex tissue for scRNA-seq^{10,11}, as well as discrete, discontinuous prenatal developmental stages in human and non-human primates^{7,12,13,14}. The diversity and features of primate cortical progenitors have been explored^{4,6,7,15}.

The temporally divergent regulatory mechanisms that govern cortical neuronal diversification at the early postmitotic stage have also been focused on¹⁶. Studies spanning the full embryonic neurogenic stage in the neocortex of humans and other primates are still lacking. Rhesus macaque and humans share multiple aspects of neurogenesis, and more importantly, the rhesus monkey and human brains share more similar gene expression patterns than the brains of mice and humans^{17–19}. To establish a comprehensive, global picture of the neurogenic processes in the rhesus macaque neocortex, which can be informative of neocortex evolution in humans, we sampled neocortical tissue at five developmental stages (E40, E50, E70, E80, and E90) in rhesus macaque embryos, spanning the full neurogenesis period. Through strict quality control, cell type annotation, and lineage trajectory inference, we identified two broad transcriptomic programs responsible for the differentiation of deep-layer and upper-layer neurons. We also defined the temporal expression patterns of neural stem cells, including oRGs, vRGs, and IPs, and identified novel transcription factors involved in oRG generation. These findings can substantially enhance our understanding of neocortical development and evolution in primates.

Results

scRNA-Seq Analysis of Cell Types in the Developing Macaque Neocortex

In order to establish a comprehensive view of the cellular composition of the rhesus macaque brain at different stages in prenatal period development, we conducted single-cell RNA sequencing (scRNA-seq) on the dissected parietal lobes of eight total rhesus macaque embryos, spanning five developmental stages of prenatal neurogenesis, including E40 (stage of peak neurogenesis) and E50 (Layer 6 formation); as well as at E70 (Layer 5 formation), E80 (Layer 4 formation), and E90 (Layer 2–3 formation) (Figure 1A, Figure S1A)². We obtained a transcriptome from 53,295 cells after filtering out low-quality cells and removing potential doublets. Each embedding was visualized using uniform manifold approximation and projection (UMAP) of dimension reduction using Seurat, which identified 28 distinct clusters. All cell clusters present in the samples were then annotated (Figure 1B), and the respective cell types were identified (Figure 1C) based on their expression of molecular markers (Figure 1D and Figure S2, B to C). According to the expression of the marker genes, we assigned clusters to cell type identities of neurocytes (including radial glia (RG), outer radial glia (oRG), intermediate progenitor cells (IPCs), ventral precursor cells (VP), excitatory neurons (EN), inhibitory neurons (IN), oligodendrocyte progenitor cells (OPC), oligodendrocytes, astrocytes, ventral LGE-derived interneuron precursors and Cajal-Retzius cells, or non-neuronal cell types (including microglia, endothelial, meninge/VALC(vascular cell)/pericyte, and blood cells). Based on the expression of the marker gene, cluster 23 was identified as thalamic cells, which are small numbers of non-cortical cells captured in the sample collection at earlier time points. Each cell cluster was composed of multiple embryo samples, and the samples from similar stages generally harbored similar distributions of cell types.

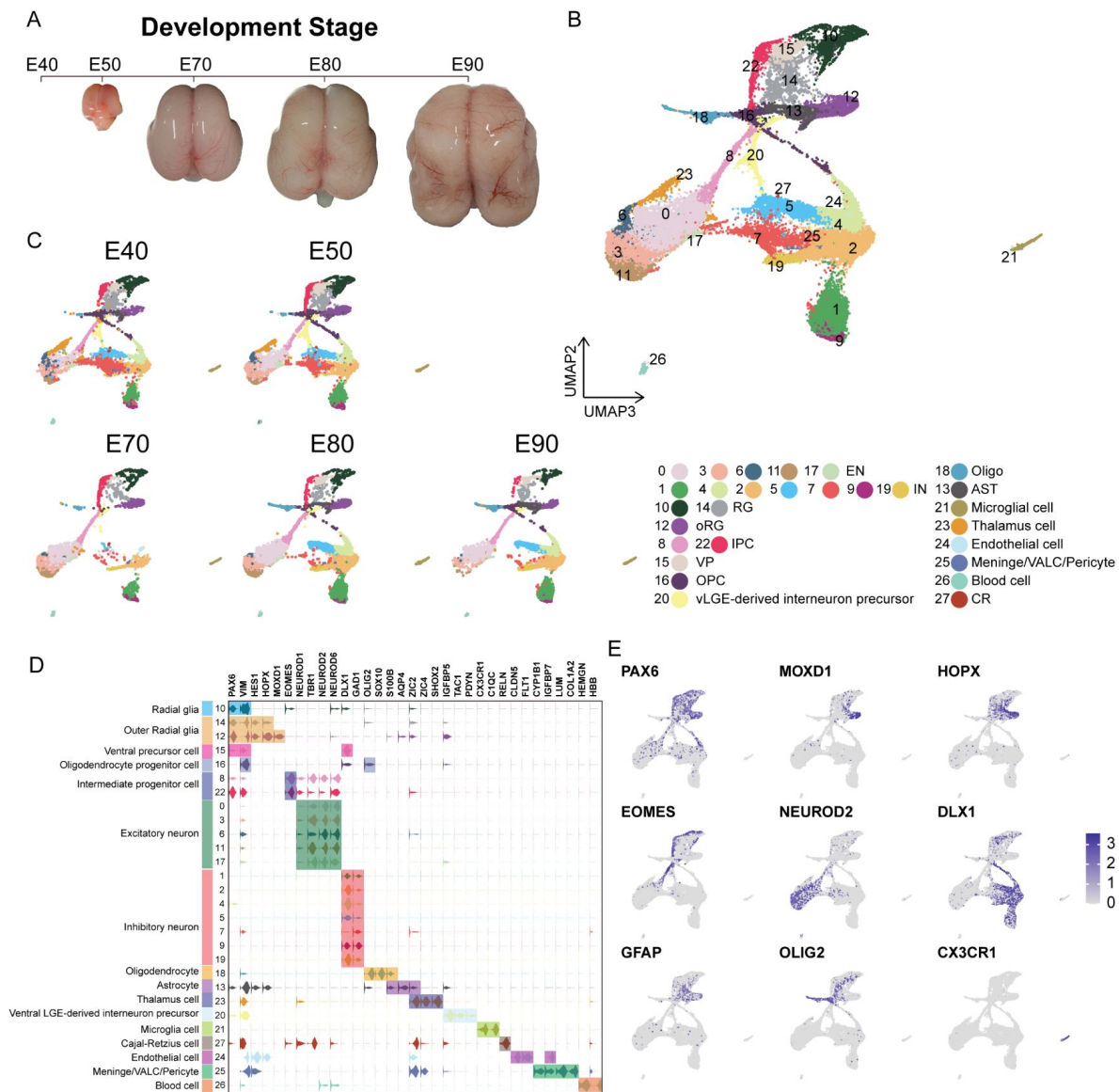


Figure 1.

Cell types in macaque prenatal and fetal brain development.

(A) Schematic diagram of sample collecting and data analysis. We collected the parietal lobe from the embryos across developmental stages from E40 to E90. (B) and (C) The transcriptome data of single cells were collected and used to do clustering using Seurat. Visualization of major types of cells using UMAP. Dots, individual cells; color, clusters. (D) Violin plot of molecular markers for annotating cell types. (E) The expressions of the classic marker genes for each cell type were plotted for UMAP visualization. Light grey, no expression; Dark blue, relative expression.

In general, ventricular radial glia (vRG) (cluster 10) showed characteristic *VIM* and *PAX6* expression; outer radial glia (oRG) (cluster 12 and cluster 14) highly expressed *HOPX*, previously verified marker⁷; two clusters (cluster 8 and cluster 22) of intermediate progenitor cells (IPCs) that strongly expressed *EOMES*, and one cluster of IPCs (cluster 22) were all topologically close to radial glia, while the other IPC cluster (cluster 8) was closer to neurons, indicating the presence of two stages of IPCs²⁰; excitatory neurons were identified by the expression of well-established markers, such as *NEUROD2* and *NEUROD6*; and the astrocyte and oligodendrocyte lineages were identified by *AQP4* and *SOX10* expression, respectively²¹. In addition, we identified *DLX1* and *GAD*-positive cells, which suggested the presence of inhibitory neurons (**Figure 1C** and **Figure S2**).

Collectively, these results suggested that cortical neural progenitors undergo neurogenesis processes during the early stages of macaque prenatal cortical development, while gliogenic differentiation, including oligodendrocytes and astrocytes, occurs in later stages (**Figure 1A** and **1B**).

Distinct Excitatory Neuronal Types Sequentially Emerge in Developing Cortex

To better understand the temporal dynamics of excitatory neuron (EN) development and differentiation, we compiled a subset of all ENs and re-clustered them into ten subclusters (EN1-10) (**Figure 2A**). We then calculated the relative expression levels of cellular markers for each group and annotated the EN subclusters based on published descriptions of marker function (**Figure S3B**). We found that all EN subclusters could be well-distinguished by differential expression of either deep-layer neuron markers (*BCL11B*, *FEZF2*, and *SOX5*)²² or upper-layer neuron markers (*CUX1* and *SATB2*)^{23,24} (**Figure 2B to C**). In previous seminal studies, 3H-thymidine (3H-dT) tracing in macaque rhesus unraveled the sequential generation of cortical neurons, with deep-layer neurons appearing prior to upper-layer neurons². In agreement with this early work, we found that early-born neurons (E40 and E50) predominantly outnumbered later-born neurons in the deep-layer neuron subclusters (EN5 and 10), while upper-layer neuron subclusters contained the inverse proportions (**Figure 2C, C and D**).

We then characterized temporal changes in the composition of each EN subcluster. While the EN 5 and EN 10 (deep-layer neurons) subclusters emerged at E40 and E50 and disappeared in later stages, EN subclusters 1, 2, 3, and 4 gradually increased in population size from E50 to E80, EN subclusters 8 and 9 gradually increased in population size from E80 to E90 (**Figure 2D**). Notably, EN 6 was exclusively found in the cortex at E70 (**Figure S3A**). EN7 is identified as *CUX1*-positive, *PBX3*-positive, and *ZFH3*-positive excitatory neuron subcluster.

Specification of Different Progenitor Fates Controlled by Regulator Genes

To focus on the differentiation of neural progenitors, we generated subsets of cell clusters 8, 10, 12, 14, 15, 16, and 22, then annotated them as ventricular radial glia (RG_C10), outer radial glia (oRG_C12 and oRG_C14), intermediate progenitor cells (IPC_C22 and IPC_C8), ventral precursor cells (VP_C15), or oligodendrocyte precursor cells (OPC_C16) (**Figure 3A**). Subclustering analysis revealed that these neural progenitors differentiated into diverse cell types with distinct lineages across the macaque prenatal neocortex (**Figure 3B**). We next investigated trajectories within the neural stem cell pool prior to branching as upper-layer or deep-layer excitatory neurons (EN). The oRG and IPC precursor cell groups exhibited characteristically high specific expression of *HOPX* and *EOMES* respectively.

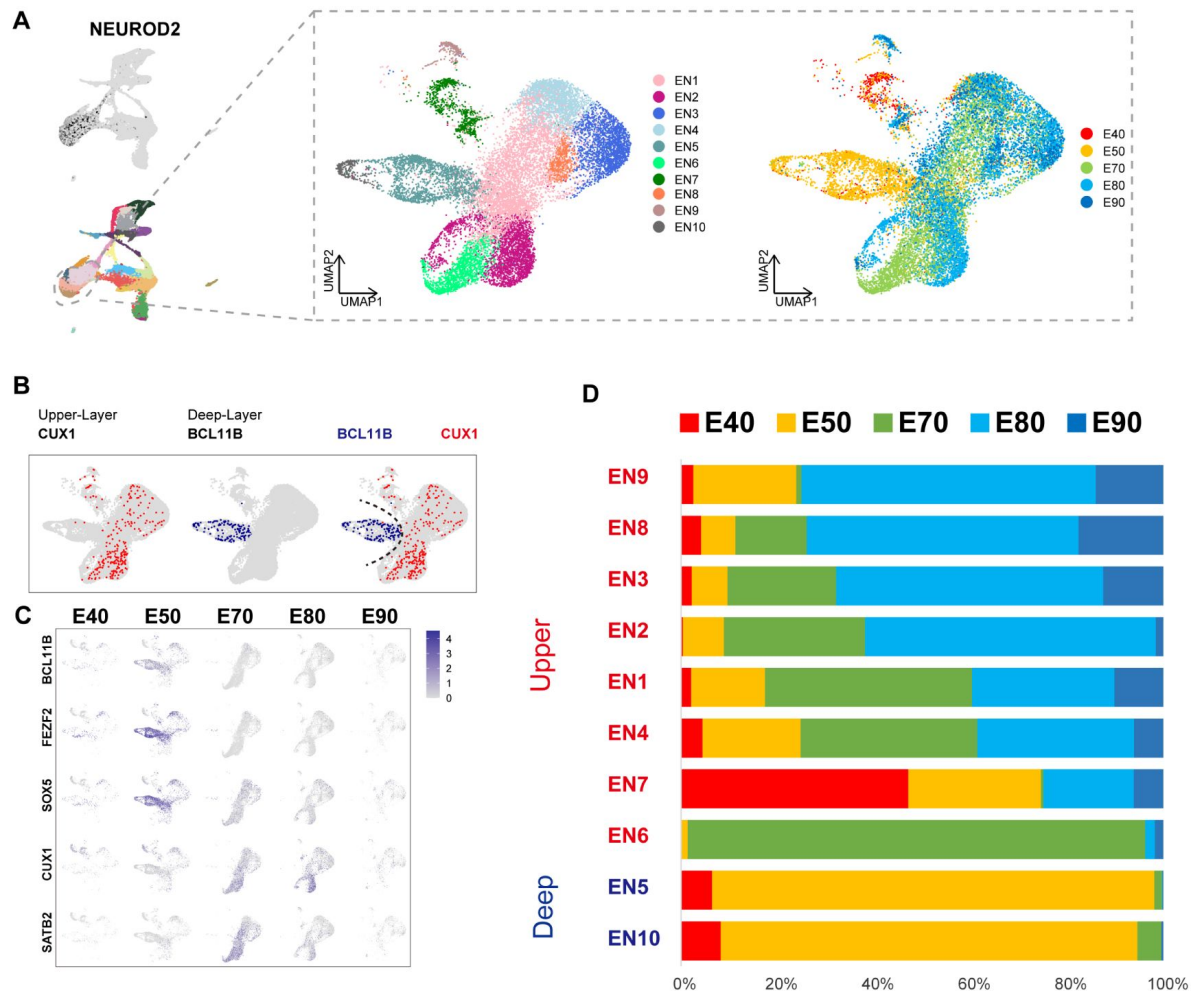


Figure 2.

Excitatory neuron subclusters in the developing macaque cerebral cortex.

(A) Left, Clustering of excitatory neuron subclusters collected at all time points, visualized via UMAP. Cells are colored according to subcluster identities (left) and collection time points (right). (B) Differentially, the expression of deep-layer marker *BCL11B* and upper-layer marker *CUX1* are highlighted. (C) Excitatory neuron subclusters UMAP plot shows the expression of classic markers for deep layers (*BCL11B*, *FEZF2*, *SOX5*) and upper layers (*CUX1*, *SATB2*) present at each time point. (D) The proportion of different excitatory neuron subclusters corresponding to excitatory neurons in each time point.

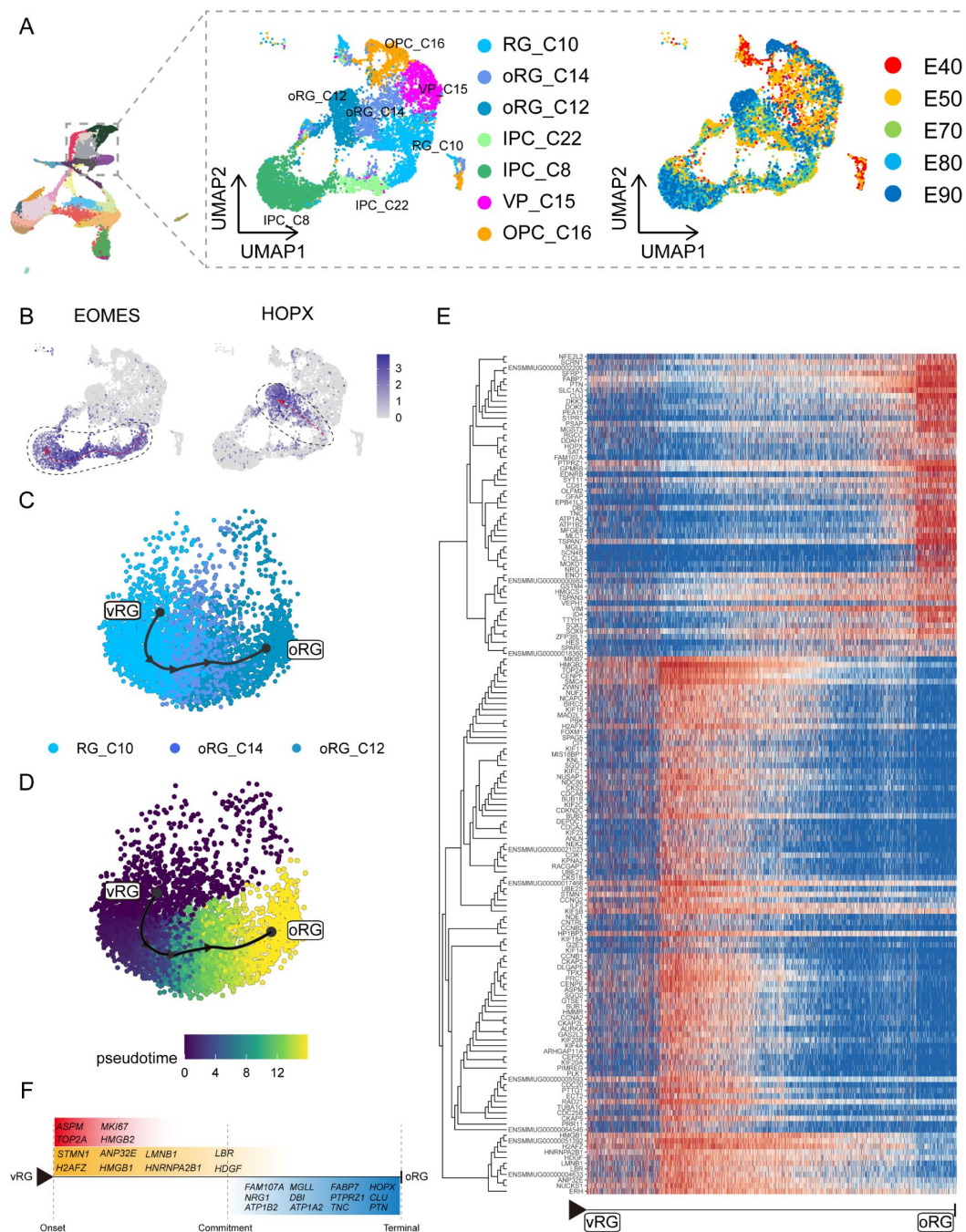


Figure 3.

Cell diversity and regulation of progenitor cells in the macaque cortical neurogenesis.

(A) UMAP shows eight progenitor clusters and cell annotation. Left cells are colored according to Seurat clusters; right, cells are colored according to the collection time point. (B) Feature plot of outer radial glia marker genes HOPX shows higher expression in C10-C14-C12 (left). Feature plots of Intermediate progenitor cell marker gene EOMES show higher expression in C10-C22-C8 (right). (C) and (D) Pseudotime analysis by Slingshot of HOPX-positive cells (C10-C14-C12). The Slingshot result with the lines indicating the trajectories of lineages and the arrows indicating directions of the pseudotime. Cells are colored according to cell (C) and pseudotime (D). Dots: single cells; colors: cluster and subcluster identity. (E) The heatmap shows the relative expression of top 150 genes displaying significant changes along the pseudotime axis of RG to oRG (C10-C14-C12). The columns represent the cells being ordered along the pseudotime axis. (F) Schema diagram of some significant genes related to E. (The depth of the color indicates the levels of gene expression)

Although the presence of oRGs is a well-established feature of primate neurogenesis, and their molecular markers are widely used, the genetic basis and molecular processes leading to their emergence are still poorly understood. To construct gene expression profiles that illustrate the progression from RG_C10 cells to oRGs, we specifically examined changes in expression among RG_C10, oRG_C12, and oRG_C14 cells, then calculated pseudotime trajectories (**Figure 3** [↗](#), **C to D**). Based on these trajectories, we then profiled the temporal shifts in the expression of each gene and selected the 150 most significant genes (**Figure 3E** [↗](#)). We found a set of genes that were previously reported as highly expressed in outer radial glia that also showed high expression in the oRG_C12 differentiated pseudotime terminal [7](#) [↗](#), [25](#) [↗](#), [26](#) [↗](#) (such as *SFRP1*, *HOPX*, *FAM107A*, *TNC*, *PTN*, and *MOXD1*), while high *ASPM* expression was typical of cells in the earlier, RG, pseudotime terminal (**Figure 3F** [↗](#)). In addition to these markers, we also found enrichment for some potential regulatory genes at the oRG_C12 terminal, such as Regulator of the cell cycle (*RGCC*), which controls mitotic spindle orientation [27](#) [↗](#), and *TTYH1*, which regulates cell adhesion [28](#) [↗](#). We also found genes that regulate ion channel expression, such as *ATP1A2*, *ATP1B2*, and *SCN4B*, which were not previously known to participate in oRG differentiation.

Analysis of DEGs specific to the RG pseudotime terminal identified cell cycle-related genes such as *MKI67*, *TOP2A*, *CDC20*, and *CCNA2*. Within the glia-like cells that largely comprised the oRG_C12 terminal, we also detected a population of true glial genes that exhibited high expression of *SLC1A3*, *ZFP36L1* [29](#) [↗](#), *GFAP*, *DBI*, and *EDNRB* [30](#) [↗](#). In addition, our analyses uncovered several DEGs that have not yet been investigated for a role in radial glia function and differentiation, such as *DKK3*, *DDAH1*, *SAT1*, and *PEA15*. Finally, we screened for significant DEGs associated with RG-to-IPC and OPC-to-astrocyte/oligodendrocyte differentiation trajectories (Figures **S5** and **S6**).

The Generation of Deep-layer and Upper-layer Neurons Follows Distinct Terminal Trajectories

Based on our finding above that the deep-layer neurons appear earlier than upper-layer neurons, we next sought to characterize dynamic shifts in the expression of transcriptional regulators that potentially contribute to determining neuronal fates. To this end, we performed pseudotime trajectory analysis of EN lineages (including deep-layer and upper-layer neurons) (**Figure 4A** [↗](#)), then superimposed the cluster labels from RG_C10, oRG_C12, and oRG_C14 stem cells in UMAP plots (**Figure 3A** [↗](#)) and early (E40 and E50) and late (E70, 80 and 90) emergence stages (**Figure 2A** [↗](#)) over the extracted transcriptomic data. We then calculated the RG-specific differentially expressed genes (rgDEGs) and the DEGs of early and late neuron-specific (nDEGs). The set of genes that overlapped between rgDEGs and nDEGs (termed mapping genes) can thus be used to map neuronal subtypes to different neural stem cell populations (**Figure. S4**). We hypothesized that mapping genes, such as *FEZF2* and *DOK5*, may function in radial glial cells to specify neuronal progeny (**Figure 4B** [↗](#)). Our results provide single-cell transcriptome data to support that apical progenitors and daughter excitatory neurons share molecular temporal identities during neocortex prenatal corticogenesis [31](#) [↗](#)–[33](#) [↗](#).

Since substantial evidence indicates that neuronal fate is determined post-mitotically in the mammalian neocortex, analysis of differentially expressed genes throughout the differentiation process could reveal the genetic mechanisms responsible for deciding stem cell fate as different neuronal progeny. Using Dynverse R packages, we constructed pseudotime trajectories for the cluster data, which identified a bifurcating trajectory from neural stem cells (including RG_C10, oRG_C12, and oRG_C14) that leads to either deep-layer neurons in one branch or to upper-layer neurons in the alternative branch (**Figure 4C** [↗](#)). However, it should be noted that the two distinct fates share a common path from neural stem cells to immature neurons (**Figure 4C** [↗](#), from **dot1** to **dot4**), supporting the likelihood that neuronal fate is primarily determined post-mitotically.

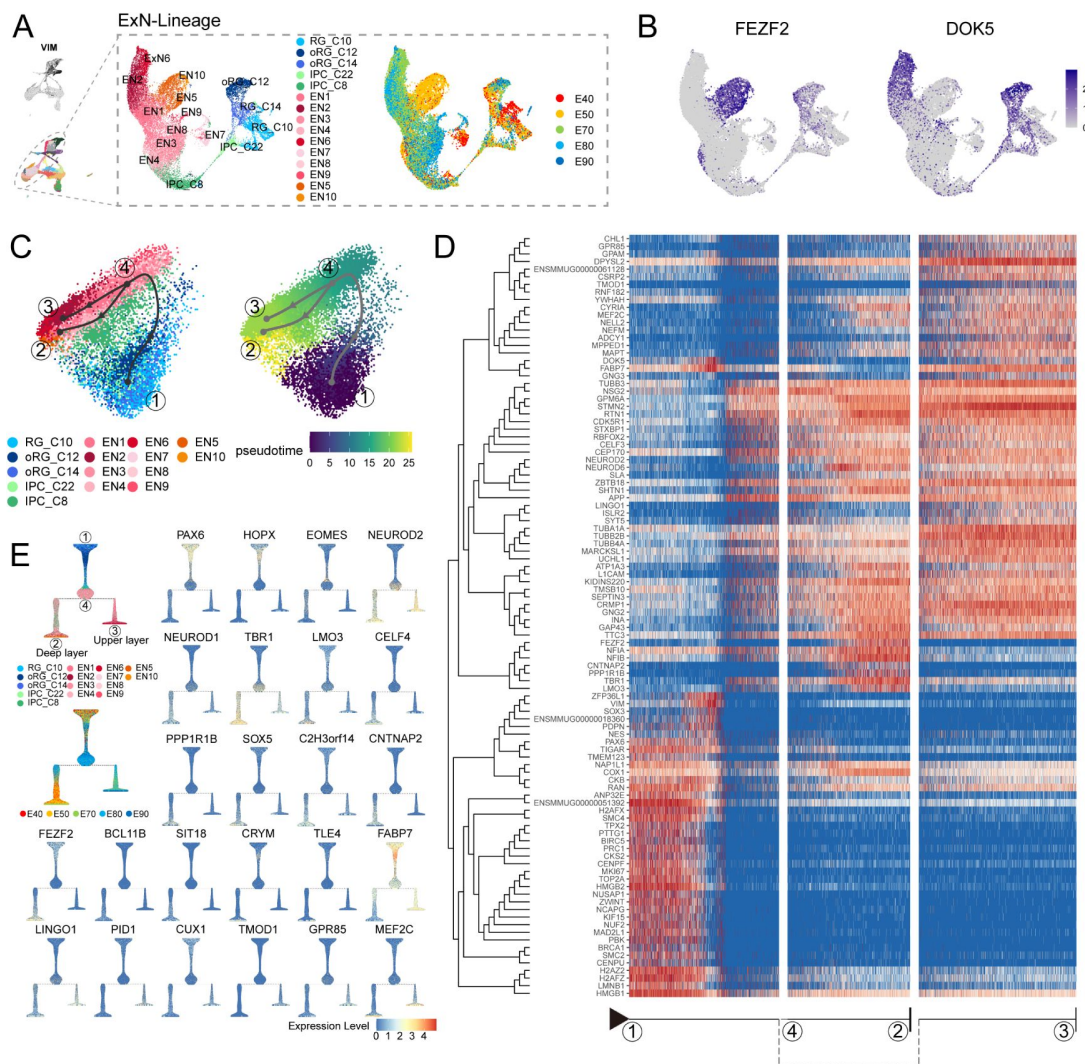


Figure 4.

Transcriptional regulation of excitatory neuron lineage during prenatal cortical neurogenesis.

(A) UMAP shows the alignment of macaque cortical NPCs, IPCs, and excitatory neurons. Left, cells are colored according to cell annotation. Different yellow/orange colors are used for deep-layer excitatory neuron subclusters (EN5 and EN10), and different red/pink colors are used for upper-layer excitatory neuron subclusters (EN1, EN2, EN3, EN4, EN6, EN7, EN9 and EN10). Right, cells are colored according to the time point of collection. (B) Dot plot showing the marker genes for the deep-layer excitatory neuron (*FEZF2*) and upper-layer excitatory neuron (*DOK5*). Light grey, no expression; Dark blue, relative expression. (C) Pseudotime analysis by Slingshot projected on PCA plot of RGCs, oRGCs, IPCs, and excitatory neuron subclusters. The Slingshot result indicates the trajectories of lineages, and the arrows indicate the directions of the pseudotime. Dots: single cell; colors: cluster and subcluster identity. Framed numbers marked the start point, endpoint, and essential nodes of the Slingshot inference trajectory. Framed number "1" was the excitatory neuron lineage trajectory start point (C10). Framed number "4" marked immature neurons. Framed numbers "2" and "3" marked deep-layer and upper-layer neurons. Cells are colored according to cell annotation and pseudotime. (D) The heatmap shows the relative expression of the top 100 genes displaying significant changes along the pseudotime axis of each lineage branch. The columns represent the cells being ordered along the pseudotime axis. (E) Left, Slingshot branching tree related to Slingshot pseudotime analysis in C. The root is E40 earliest RG (C10), tips are deep-layer excitatory neurons generated at the early stage (E40, E50), and upper-layer excitatory neurons are generated at the later stage (E70, E80, E90). Right, branching trees showing the expression of marker genes of apical progenitors (*PAX6*), outer radial glia cells (*HOPX*), intermediate progenitors (*EOMES*), and excitatory neurons (*NEUROD2*), as well as genes in Figure 3E, including callosal neurons (*SATB2*, *CUX2*), deeper layer neurons (*SOX5*, *FEZF2*), corticofugal neurons (*FEZF2*, *TLE4*). There is a sequential progression of radial glia cells, intermediate progenitors, and excitatory neurons.

Based on this trajectory, we categorized the DEGs that exhibit dynamic, temporal shifts in expression over pseudotime, resulting in broad clusters. The first of these clusters was enriched with neural stem cells, characterized by high expression of *PAX6*, *VIM*, and *TOP2A*, which likely participate in regulating stemness and proliferation. The three remaining DEG clusters were enriched in either deep-layer neurons, upper-layer neurons, or both (**Figure 4D** [↗](#)). More specifically, we found that most DEGs, such as *STMN2*, *TUBB3*, *NEUROD2*, and *NEUROD6*, were shared between both branches, illustrating the differentiation processes between deep and upper-layer ENs. We also identified deep layer-specific DEGs (including *FEZF2*, *TBR1*, and *LMO3*) and upper layer-specific genes (including *MEF2C* and *DOK5*). We then organized the cell types into a branching tree based on their differential expression of these marker genes, including stem-cell genes *PAX6*, *HOPX*, and *EOMES*, in addition to the above deep-layer-and upper-layer-specific genes (**Figure 4E** [↗](#)). The separation into distinct, branch-specific sets of DEGs suggested a role for these cell type-specific transcription factors and other genes in terminal neuron differentiation.

Conserved and Divergent Features of Human, Macaque and Mouse Neocortical Progenitor Cells

Next, we performed liger integration ³⁴[↗](#) to integrate the macaque single-cell dataset with published mouse ¹¹[↗](#) and human datasets, which were created by combing two published human scRNA-seq data of 7 to 21 postconceptional week (PCW) neocortex ^{35,36}[↗](#), to conduct a cross-species comparison. Species-integrated UMAP showed that all major cell types were well integrated between the three species (**Figure 5A** [↗](#)).

It is clearly observed that similar cellular processes with comparable temporal progression sequentially generate deep-layer and upper-layer neurons in both primates and rodents (**Figure S7, A to C**). However, in primates, the number and relative development period of upper neurons are more extended than that of rodents (**Figure S7D**). Besides, another important evolutionarily divergent feature in cortical development between primates and rodents is the abundance of oRG cells in primates' outer subventricular zone (oSVZ) (human and macaque). Our human-macaque-mouse cross-species analysis showed that strong expression of the oRG marker genes, such as *HOPX*, *MOXD1*, *FAM107A* and *CLU* (3) in both human and macaque datasets (**Figure 5B** [↗](#)), while barely detectable expression in mouse dataset, which is consistent with previous reports ³⁷[↗](#).

Furthermore, We picked *HOPX*-positive and *EOMES*-positive cells (ligercluster 5, 20, 13, 15) from the human, macaque, and mouse datasets and performed developmental trajectories analysis with slingshot (**Figure 5C** [↗](#)). Comparing vRG to IPC trajectory between human, macaque, and mouse, we found this biological process of vRG-to-IPC is remarkably conserved across species. However, the vRG to oRG trajectory is divergent between species because the oRG population was not identified in the mouse dataset. The latter process is almost invisible in mice but similar in humans and macaques.

Previous studies have shown that neural progenitors exhibit distinct temporal expression patterns during neurogenesis in mice. However, similar temporal profiling has yet to be thoroughly investigated in macaque neural stem cells. Based on its topological position and associated markers that suggest it functions as a developmental root, we examined the temporal expression profile of RG_C10 cells (**Figure 3A** [↗](#)). We first identified marker genes to distinguish RG_C10 cells in different prenatal stages (i.e., E40-E90) and used these markers to generate dynamic expression profiles. We then clustered these dynamically expressed macaque genes into five types (**Figure 6A** [↗](#)), consisting of Type 1 and 2 genes that decreased in expression during neurogenesis, Type 3, 4 and 5 genes that gradually ramped up in expression throughout neurogenesis. We then performed a similar profiling of dynamic gene expression using data obtained from mouse apical progenitors¹⁰[↗](#) and categorized the DEGs into the same five categories. A total of 72 homologous transcription factors were shared between human, macaque and mouse transcriptomes. Analysis of their temporal dynamics revealed that they exhibited similar temporal patterns between

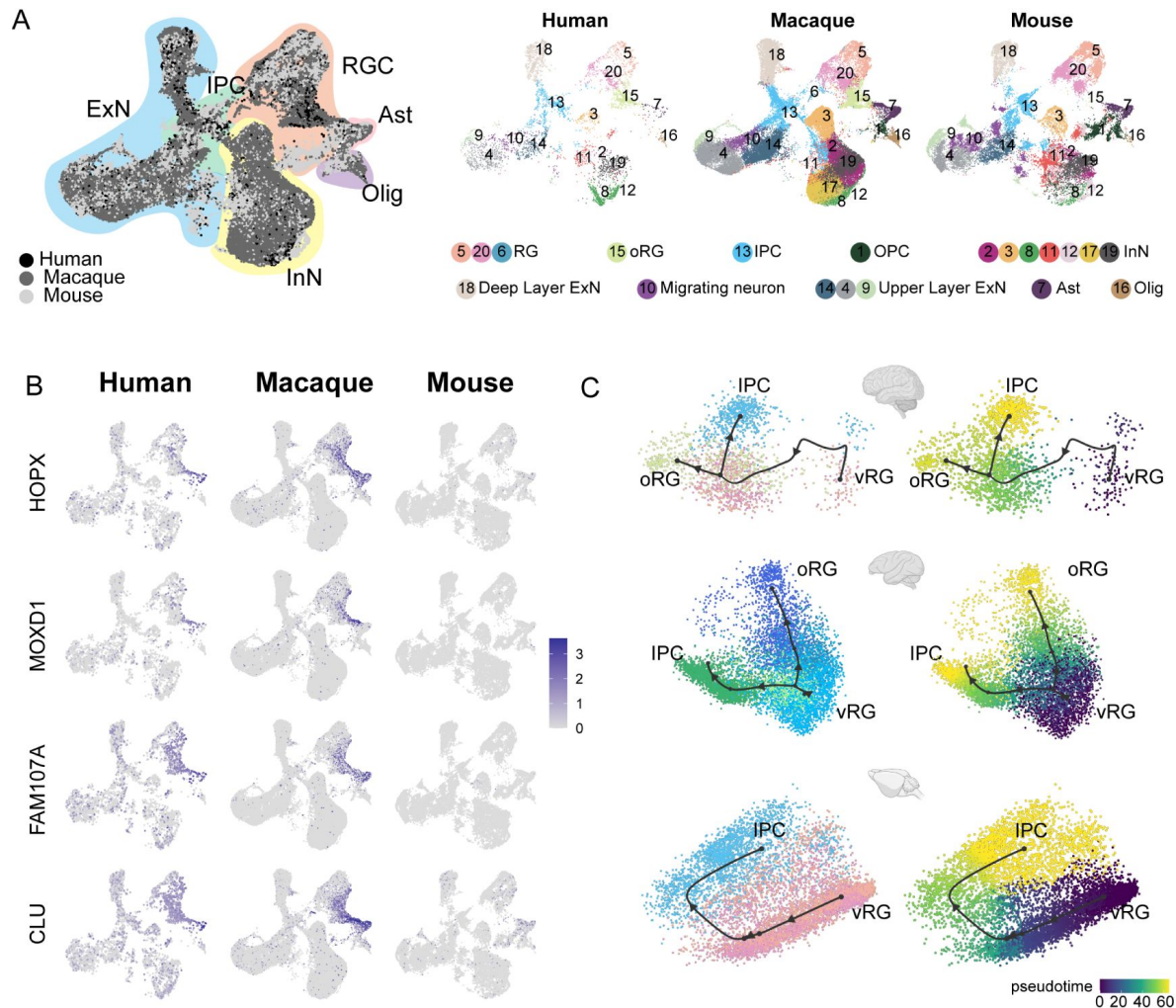


Figure 5.

Integration of human, macaque and mouse single-cell datasets reveals conserved and divergent progenitor cell types.

(A) Left: UMAP plot of cross-species integrated single-cell transcriptome data with liger. Colors represent different major cell types (Black: Human dataset; Dark grey: macaque dataset; Liger grey: mouse). Right: The UMAP plot of each dataset, colored by liger cluster. (B) The expressions of the classic oRG marker genes were plotted to UMAP visualization. Light grey, no expression; Dark blue, relative expression. (C) Comparison of vRG → oRG and vRG → IPC developmental trajectories between human, macaque, and mouse.

species such as *NEUROD1*, *NEUROG2*, *POU3F2*, *ETV1*, *ETV5* and *FOS* (**Figure 6B** [↗](#) and **Figure S8**), which indicated that the majority of transcription factors had conserved temporal dynamics between species, and thus evolutionarily conserved patterns of regulation in neural stem cell differentiation.

To identify the master regulators (MRs) related to cortical neurogenesis among macaque and mouse, we used the SCENIC workflow to analyze the gene regulatory networks of each transcription factor (**Figure 6** [↗](#), **C to H**). Among all the regulatory pairs associated with each prenatal stage, we selected ten TFs with the highest regulon specificity scores (**Figure 6** [↗](#) **C, E and G**) and their top 5 target genes, as visualized by Cytoscape. Analysis of the regulatory activity of human, macaque and mouse¹¹ [↗](#) prenatal neocortical neurogenesis indicated that despite commonalities in the roles of classical developmental TFs such as *GATA1*, *SOX2*, *HMGN3*, *TCF7L1*, *ZFX*, *EMX2*, *SOX10*, *NEUROG1*, *NEUROD1* and *POU3F1*. The top 10 TFs of the human, macaque, and mouse vRG each time point and their top 5 target genes identified by pySCENIC as an input to construct the transcriptional regulation network (**Figure 6 D, F** [↗](#) and **H**). Some conserved regulatory TFs present in more than one species are identified, such as *HMGN3*, *EMX2*, *SOX2*, and *HMG2* at an early stage when deep-layer generation and *SOX10*, *ZNF672*, and *ZNF672* at a late stage when upper-layer generation.

Discussion

The cerebral cortex region of the brain is responsible for extraordinary cognitive capacities, such as abstract thinking and language. The neocortex of non-human primate rhesus macaque resembles that of humans in many aspects. Thus, a comprehensive investigation of macaque neurogenesis can improve our understanding of neocortical development and evolution. For this purpose, we performed scRNA-seq in prenatal neocortical tissues of macaque. Our data support previous omics studies of prenatal macaque brain development³⁸ [↗](#) ³⁹ [↗](#) ¹⁷ [↗](#) and recently published human neocortex development studies⁴⁰ [↗](#). We constructed a single-cell resolution transcriptomic atlas of the developing macaque neocortex, which we then used to identify dynamically expressed genes that likely contribute to the maturation of distinct neuron types, temporal expression patterns of neural stem cells, and the generation of oRGs.

The existence of oRGs has long been demonstrated in humans and primates through live imaging and immunostaining⁴¹ [↗](#). ORGs share similarities in their patterns of gene expression with vRGs (e.g., *SOX2*, *PAX6*, and *NESTIN*) but also specifically express several genes (such as *MOXD1*, *HOPX*, and *FAM107A*)⁷ [↗](#), ⁴² [↗](#). Recently, the molecular mechanisms associated with the generation and amplification of oRGs were shown to include signaling pathways such as the *FGF-MAPK* cascade, *SHH*, *PTEN/AKT*, and *PDGF* pathways, as well as proteins such as *INSM*, *GPSM2*, *ASPM*, *TRNP1*, *ARHGAP11B*, *PAX6*, and *HIF1α*. A number of these proteins were validated by genetic manipulation of their activities or expression in mouse, ferret, and marmoset⁴³ [↗](#)–⁴⁶ [↗](#). The dynamically expressed genes and pseudotime trajectories from vRGs to oRGs presented in this work are in agreement with these previous reports. In addition to known regulators, we detected differential expression of ion channel regulators in oRGs, such as *ATP1A2*, *ATP1B2*, and *SCN4B*, suggesting that hyperpolarization and depolarization processes may participate in promoting or maintaining oRG populations. Indeed, hyperpolarization in mouse apical progenitors has been shown to promote the generation of IPCs and indirect neurogenesis⁴⁷ [↗](#), although it remains unclear if similar mechanisms contribute to the development of the primate neocortex.

The sequential generation of cortical neurons has long been observed in both rodents and primates, and many transcription factors are known to drive post-mitotic specification of neuronal progeny, mainly in rodents. Here, we found genes conserved across species, such as *FEZF2* and *TBR1*, involved in specifying deep-layer neurons in the macaque neocortex. The slingshot pseudotime analysis, which reflected the expression pattern of genes during appropriate lineage

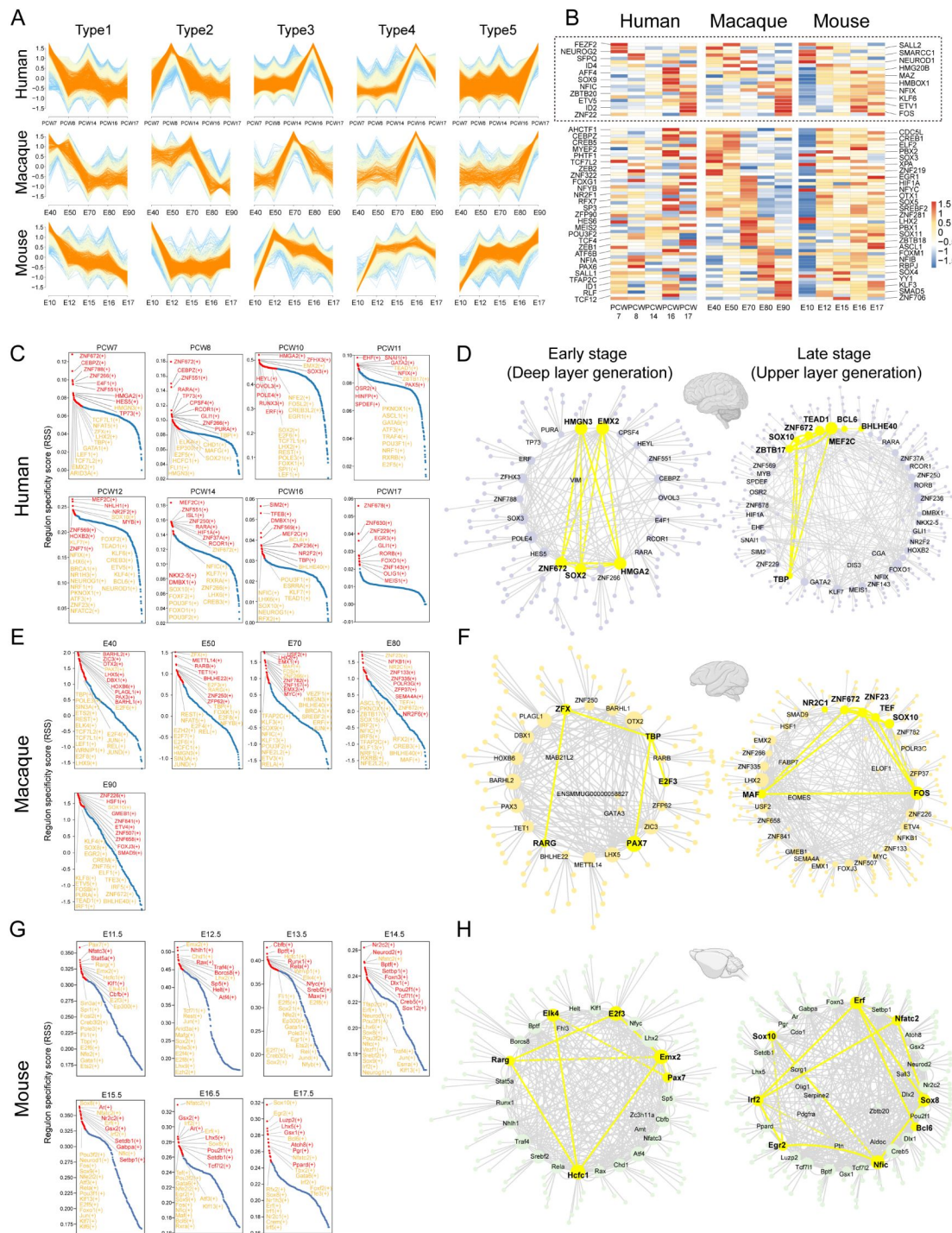


Figure 6.

The patterns of transcriptional regulation comparative analysis responsible in vRGs.

(A) Normalized expressions of genes that show temporal dynamics in the vRGs of human, macaque and mouse. (B) Temporal expression heatmap of homologous transcription factor genes among human, macaque, and mouse temporal expression heatmap. (The TFs genes in the dashed boxes showed similar temporal expression patterns across species. (C), (E) and (G) Region specificity score for each timepoint in human, macaque, and mouse vRG. Regulons with high scores in multiple species vRG cells are colored yellow. (D), (F) and (H) show a network generated with Cytoscape using the top 10 regulons in the human, macaque, and mouse vRG at each time point and their top 5 target genes identified by SCENIC as an input. The interactions between conserved TFs in more than one species are colored yellow.

trajectories, suggests that some genes are lineage-restricted or even involved in lineage fate determination. For example, *FEZF2*, *PPP1R1B* (also known as *DARPP32*), *SOX5*, *LMO3*, and *CELF4* may be involved in the fate specialization of deep neurons. In contrast, *TMOD1*, *PID1*, *LINGO1*, *CRYM*, and *MEF2C* for upper-layer neurons. Notably, *FEZF2*⁴⁸ and *SOX5*⁴⁹ have been confirmed associated with deep layer neuron specification in mice, while others, such as *TMOD1*, *PID1*, and *LINGO1*, have not been reported before, and their potential functions in the fate specification of cortical projection neurons need to be further explored. To date, only a very limited set of genes have been shown to function in specifying neuronal progeny in progenitor cells, which suggests that regulation in progenitors is more complex than in neurons, where a single transcription factor can trigger a transcriptional cascade to promote the maturation of neurons. In the current study, we found two types of genes (Dynamic type 1 and 5), which have gradually altered expression patterns along with the neurogenic stage, indicating a transition in cell state in both vRGs and oRGs. In addition to these gradual transition genes, we also identified sets of genes (Dynamic type 2-4) with sharp changes in expression during neurogenesis, which likely function in specifying different neuronal subtypes. These findings led us to postulate that a combination of biological processes and pathways change over time to coordinate gradual transitions from neural stem cells to daughter neurons.

This study also faces limitations. During the macaque prenatal neurogenesis, dorsal radial glia cells generate excitatory neurons by direct and indirect neurogenesis. In general, the number of progenitor cells gradually decreases throughout development, with progenitor cells generating deep neurons in the early stage of development and upper excitatory neurons in the later stages. We observed the disappearance of two subclusters of ENs (EN 5 and EN 10) from later-stage samples (Figure S3). This disappearance could also be explained by the development of axons and dendrites in early-born neurons, which have higher morphological complexity and greater vulnerability. Due to the fixed size of the BD Rhapsody microwells, this sing-cell capture method might be less efficient in capturing mature excitatory neurons but has a good capture effect on newborn neurons at each sampling time point. Nonetheless, this transcriptomic atlas uncovers the molecular signatures of the primary cell types and temporal shifts in gene expression of differentiating progenitor cells during neocortex layer formation, thus providing a global perspective into neocortical neurogenesis in the macaque.

Materials and Methods Animal

Animals and frozen tissue samples from prenatal rhesus macaque (*Macaca mulatta*) were provided by the National Medical Primate Center, Institute of Medical Biology, Chinese Academy of Medical Sciences. Timed pregnancy-derived biological replicate specimens were profiled at each prenatal developmental stage (E40, E50, E70, E80 and E90)⁵⁰. These time points were selected to coincide with peak periods of neurogenesis for the different layers of the neocortex based on previous studies^{50,51}. All animal procedures followed international standards and were approved in advance by the Ethics Committee on Laboratory Animals at IMBCAMS (Institute of Medical Biology, Chinese Academy of Medical Sciences).

Fetal brain sample details

We collected eight pregnancy-derived fetal brains of rhesus macaque (*Macaca mulatta*) at five prenatal developmental stages (E40, E50, E70, E80, E90) and dissected the parietal lobe cortex. Because of the different development times of rhesus monkeys, prenatal cortex size and morphology are different. To ensure that the anatomical sites of each sample are roughly the same, we use the lateral groove as a reference to collect the parietal lobe for single-cell sequencing (as indicated by bright yellow in Figure S1A) and do not make a clear distinction between the different regional parts including primary somatosensory cortex and association cortices in the process of sampling.

Cell preparation

We followed the BD Cell Preparation Guide to wash, count, and concentrate cells in preparation for use in BD Rhapsody™ System Whole transcriptome analysis.

Single-cell RNA-seq data processing

Single-cell capture was achieved by random distribution of a single-cell suspension across >200,000 microwells through a limited dilution approach. Beads with oligonucleotide barcodes were added to saturation so that a bead was paired with a cell in a microwell. The cell-lysis buffer was added so that poly-adenylated RNA molecules hybridized with the beads. Beads were collected into a single tube for reverse transcription. Upon cDNA synthesis, each cDNA molecule was tagged on the 5' end (the 3' end of a mRNA transcript) with a molecular index and cell label indicating its cell of origin. Whole transcriptome libraries were prepared using the BD Resolve single-cell whole-transcriptome amplification workflow. In brief, the second strand of cDNA was synthesized, followed by ligation of the adaptor for universal amplification. Eighteen cycles of PCR were used to amplify the adaptor-ligated cDNA products. Sequencing libraries were prepared using random priming PCR of the whole-transcriptome amplification products to enrich the 3' ends of the transcripts linked with the cell label and molecular indices. Sequencing was performed with Illumina NovaSeq 6000 according to the manufacturer's instructions. The filtered reads were aligned to the macaque reference genome file with Bowtie2 (v2.2.9). Macaque reference genome was downloaded from Ensemble database (ftp://ftp.ensembl.org/pub/release100/fasta/macaca_mulatta/dna/Macaca_mulatta.Mmul_10.dna.toplevel.fa.gz [↗](#), ftp://ftp.ensembl.org/pub/release-100/gtf/macaca_mulatta/Macaca_mulatta.Mmul_10.100.gtf.gz [↗](#)). We analyzed the sequencing data following BD official pipeline and obtained the cell-gene expression matrix file.

QC and Data analysis

To mitigate the effect of over-estimation of molecules from PCR and sequencing errors, we used the Unique Molecular Identifier (UMI) adjustment algorithms recursive substitution error correction (RSEC) and distribution-based error correction (DBEC), which were contained in the BD Rhapsody™ pipeline. Quality control was applied based on the detected gene number and the percentage of counts originating from mitochondrial RNA, ribosomal RNA, and hemoglobin gene per cell. Then, cells were filtered to retain only higher quality (with less than 7.5% mitochondrial gene counts, less than 5.5% ribosomal gene counts, and detected genes above 400 and less than 6000). Additionally, we identified cell doublets using Scrublet v0.142 with default parameters and removed doublets before data analysis. After quality control, a total of 59,045 cells remained for subsequent analysis.

To integrate cells into a shared space from different datasets for unsupervised clustering, we used SCTransform workflow in R packages Seurat⁵² [↗](#) V4.0.3 to normalize the single-cell RNA-seq data from different samples. We identified the variable features of each donor sequencing data using the “FindVariableFeatures” function. A consensus list of 2000 variable genes was then formed by detecting the most significant recovery rates genes across samples. Mitochondrial and ribosomal genes were not included in the variable gene list. Next, we used the SCTransform workflow in Seurat to normalize the single cell-seq data from different samples. During normalization, we also removed confounding sources of variation, including mitochondrial mapping and ribosomal mapping percentages. Standard RPCA workflow was used to perform the integration.

With a “resolution” of 0.5 upon running “FindClusters,” we distinguished major cell types of nerve cells and non-nerve cells in the umap according to known markers, including radial glia cell, outer radial glia cell, intermediate progenitor cell, excitatory neuron, inhibitory neuron, astrocyte, oligodendrocyte, Cajal-Retzius cell, microglia, endothelial cell, meningeal cells, and blood cells, which were distinguished on the first level. By following a similar pipeline, subclusters were identified. We finalized the resolution parameter on the FindClusters function once the cluster

numbers did not increase when we increased the resolution. Then, we checked the differentially expressed genes (DEGs) between each of the clusters using the “FindAllMarkers” and “FindMarkers” functions with `logfc.threshold=0.25`.

Construction of single-cell developmental trajectory

Single-cell developmental lineage trajectories construction and discovery of trajectory transitions were performed using Slingshot(V 2.2.0)⁵³, from the Dyno (V 0.1.2)⁵⁴ platform, with PCA dimensionality reduction plot results. The direction of the developmental trajectory was adjusted by reference to the verified relevant studies.

Cross-species transcriptome data integration and analysis

Mouse datasets were downloaded from the Gene Expression Omnibus (GEO SuperSeries GSE153164) and at the Single Cell Portal: https://singlecell.broadinstitute.org/single_cell/study/SCP1290/molecular-logic-of-cellular-diversification-in-the-mammalian-cerebral-cortex. Human datasets were download from the Gene Expression Omnibus (GEO) under the accession number GSE104276 and GSE104276. We created a human database by combining the two published human prenatal cortical development datasets of 7 to 21 postconceptional weeks (PCW) into one containing cell numbers comparable to our macaque and published mouse dataset using the single cell transcriptome analysis pipeline of R package Seurat. We used the biomaRt package to convert the gene symbols in the mouse and macaque expression matrices into their human homologs. To perform cross-species analysis, we used the LIGER^[31] method to integrate the macaque single-cell dataset with the mouse and human datasets. We ran the “optimizeALS” function in rliger to perform integrative non-negative matrix factorization(iNMF) on the scaled species-integrated dataset ($k = 20$, $\lambda = 5$, $\text{max.iter} = 30$). Species-integrated UMAP showed that all major cell types were well integrated between the three species (**Figure 5A**). We used the “calcAlignment” function in rliger to quantify how well-aligned datasets are and got the metric of 0.891231545756746, which was very close to “1”. This suggested LIGER well-integrated cross-species datasets.

Transcription Factor Regulatory Network Analysis

We inferred regulon activities for vRG cells of human, macaque and mouse using pySCENIC^[56] workflow (<https://pyscenic.readthedocs.io/en/latest/tutorial.html>) with default parameters. Since there is no transcriptional regulator database for macaques, macaques’ genome is similar to the human genome. When we analyzed the macaque dataset, we used the human transcriptional regulator database for pySCENIC analysis. GRNboost2 in scenic was used to infer gene regulatory networks based on co-expression patterns. RcisTarget⁵⁵ was used to analyze the TF-motif enrichment and direct targets among different species databases. The regulon activity for each developmental time point was identified using AUCell, and the top enriched activated TFs were ranked by $-\log_{10}(\text{p_value})$. Lastly, we evaluated the interaction networks analysis of TFs and their target genes by Cytoscape⁵⁶ (version 3.8.1).

Mfuzz Clustering

Firstly, we picked up vRG cells in each species dataset and screened the differentially expressed genes (DEGs) between adjacent development time points using the “FindMarkers” function (with `min.pct = 0.25`, `logfc.threshold = 0.25`). After separate normalization of the DEG expression matrix from different species datasets, we use the “standardise” function of mfuzz to perform standardization. The DEGs of vRG in each species were grouped into five different clusters using the Mfuzz package in R with fuzzy c-means algorithm⁵⁷.

Transcription Factor and RNA binding protein analysis

Mouse, macaque, and human transcription factor (TF) gene lists were downloaded from AnimalTFDB (<http://bioinfo.life.hust.edu.cn/AnimalTFDB/> ) . Mouse, macaque, and human RNA binding protein (RBP) gene lists were downloaded from EuRBPDB (<http://EuRBPDB.syshospital.org> ) . Heatmaps of the TFs and RBPs expression profiles were generated with normalized and standardized DEGs expression matrixes from mfuzz.

Acknowledgements

Funding: This work was supported by CAMS Innovation Fund for Medical Sciences (CIFMS, 2021-I2M-1-024, 2021-I2M-1-019) and National Science and Technology Innovation 2030 Major Program 2021ZD0200900.

Data availability

The raw sequence data reported in this paper have been deposited in the Genome Sequence Archive in National Genomics Data Center, China National Center for Bioinformation / Beijing Institute of Genomics, Chinese Academy of Sciences (GSA: CRA007718) that are publicly accessible at <https://ngdc.cncb.ac.cn/gsa>  . Codes used to analyze results in this paper are available on GitHub at <https://github.com/cheneylemo/developing-macaque-neocortex>  .

Author Contributions

X.P. and B.Q. supervised the research. P.S. and Y.C. conceived of the experimental design and bioinformatics analysis. Y.Z., Z.H. and S.L. established the macaque prenatal developmental model. W.L. performed the anatomical analysis of monkeys and related tests. L.X. and J.Z. contributed to drafting or revising the manuscript and figures. Z.Y. and L.X. performed single-cell transcriptome data analysis.

Competing Interest Statement

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Supporting Information

Dataset S1. (separate file)

Marker list of 28 cell clusters.

Dataset S2. (separate file)

Gene expression importance across the vRG to oRG (C10-C14-C12) lineage trajectory related to **Fig. 3E**  .(vRG: milestone2; oRG: milestone4)

Dataset S3. (separate file)

Gene expression importance across the RG to deep-layer neurons and upper-layer.

Dataset S4. (separate file)

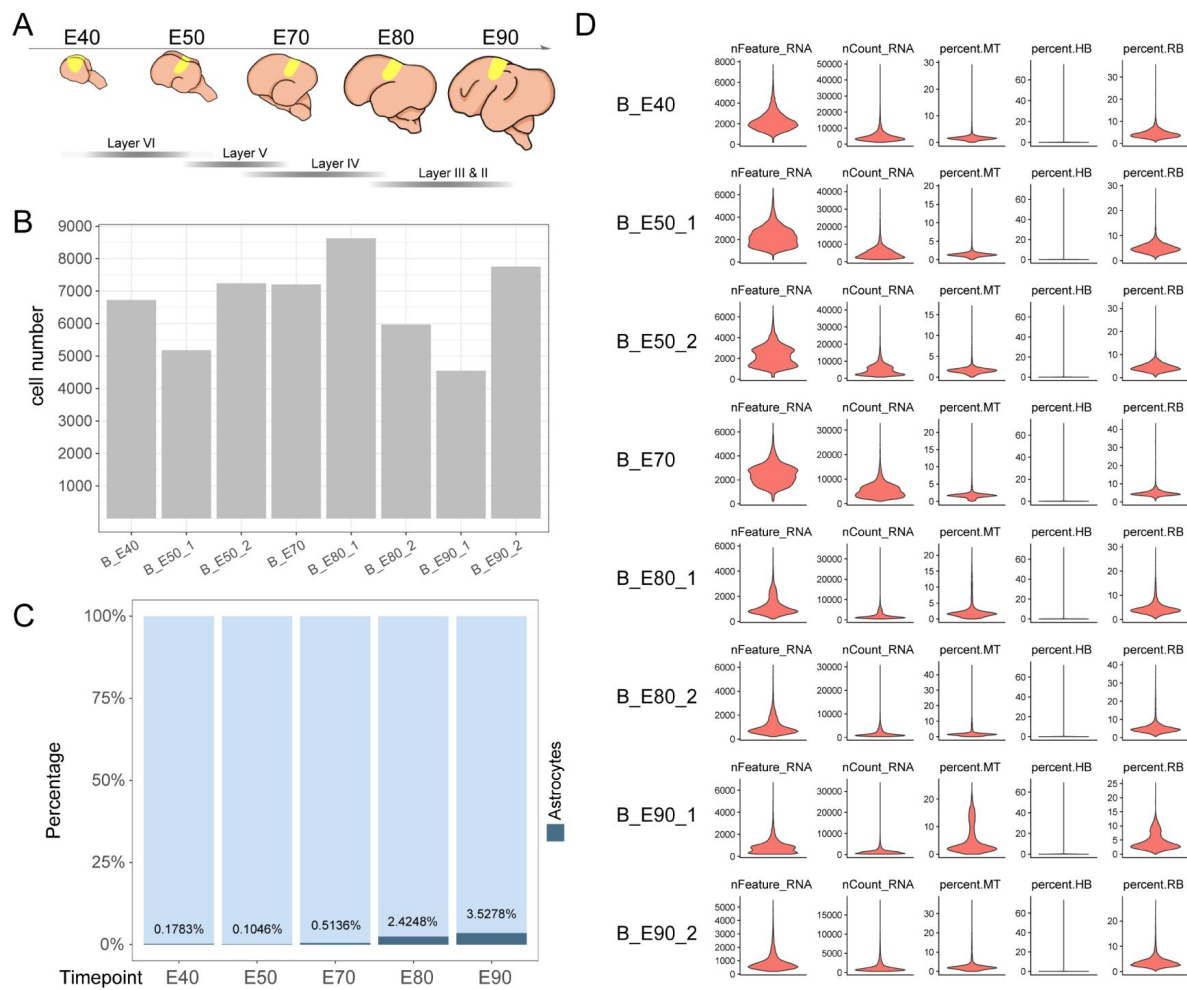


Figure S1.

Sample collection and Quality Control.

- (A) Schematic diagram of sample collecting anatomical area.
- (B) The cell number per sample after quality control.
- (C) Bar chart of astrocytes proportion statistical at each time point.
- (D) Single-cell transcriptome library information for each sample.

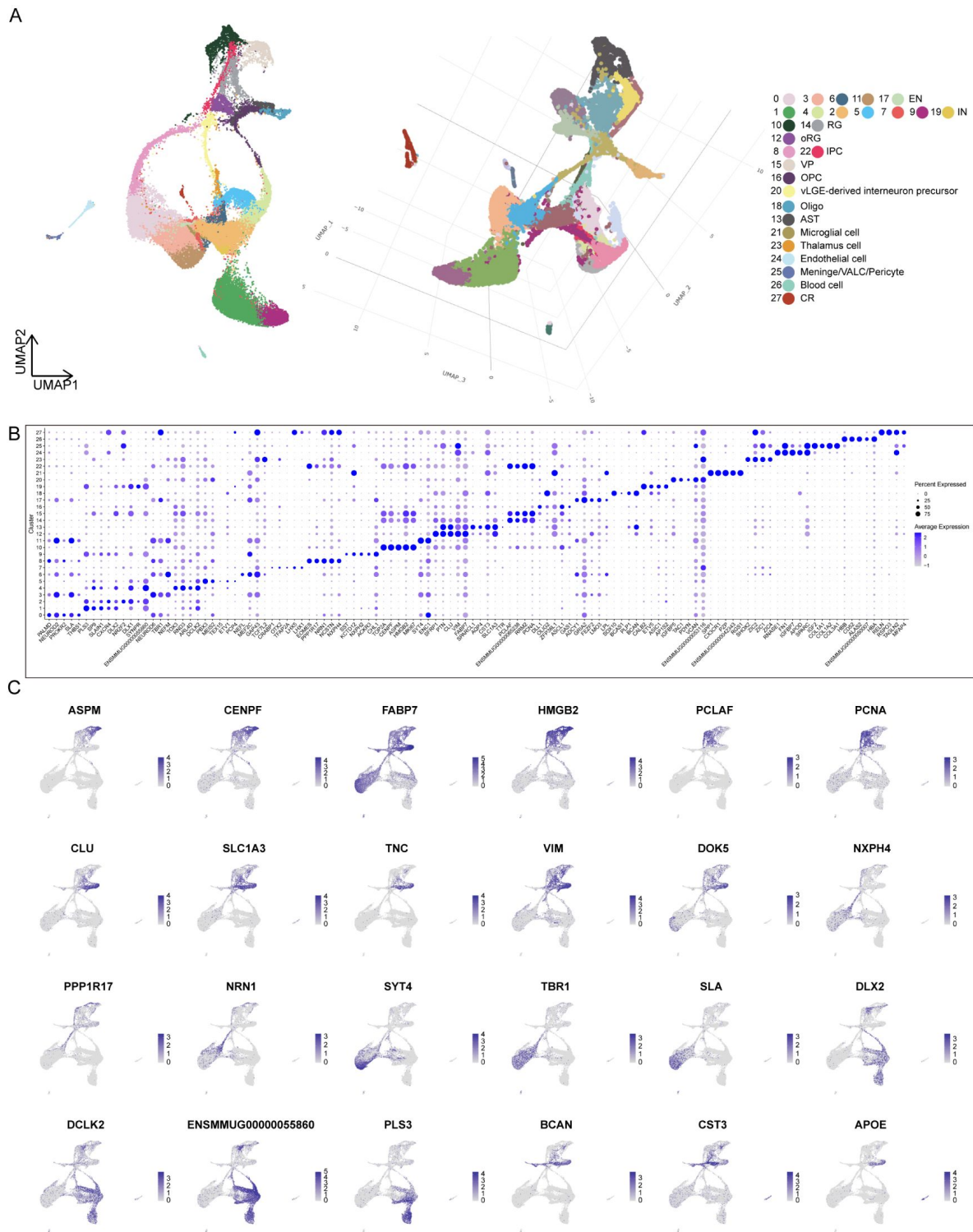


Figure S2.

scRNA-Seq uncovers cell type in the developing macaque neocortex.

(A) Visualization of different dimensionality reduction of all cells. (Left, UMAP visualization with UMAP1 and UMAP2; right, 3D model of UMAP visualization with UMAP1, UMAP2, and UMAP3).

(B) Top marker genes for each of the 28 cell clusters shown in [Figure 1B](#).

(C) Feature plots of marker gene expression. Colors represent scaled gene expression.

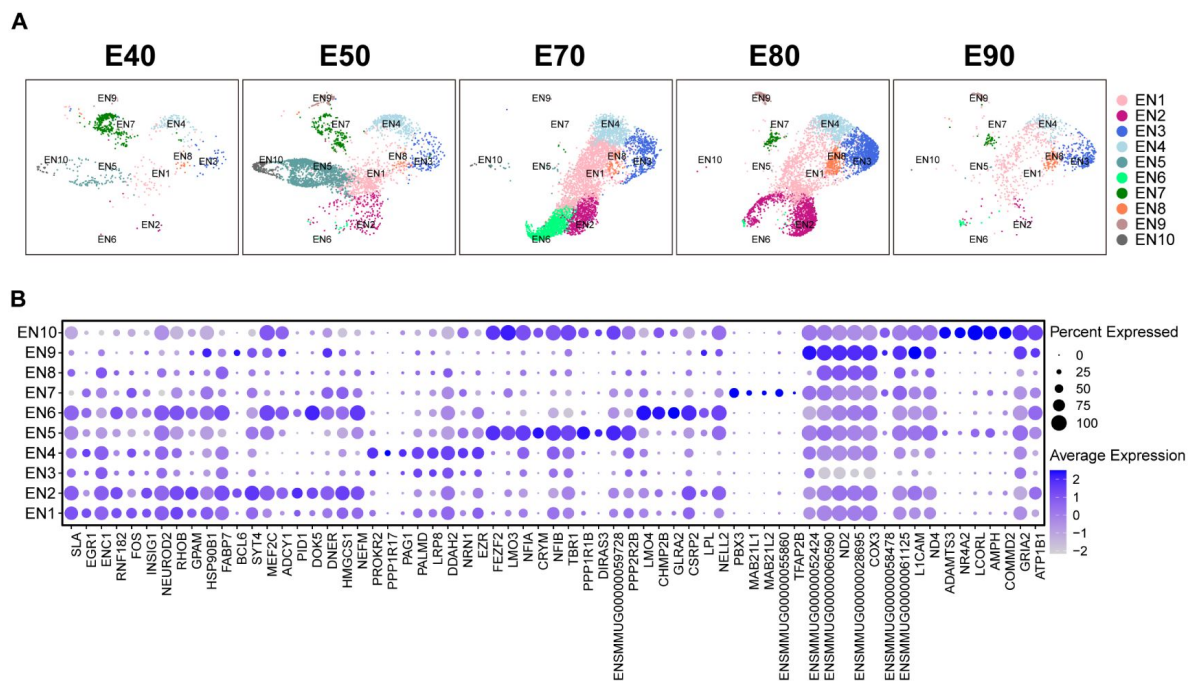


Figure S3.

Additional information for excitatory neuron subcluster.

- (A) UMAP visualization of excitatory neuron subcluster cell scRNA-seq data from individual time points. Cells are colored by excitatory neuron subcluster assignment.
- (B) Top marker genes for each of the excitatory neuron subclusters.

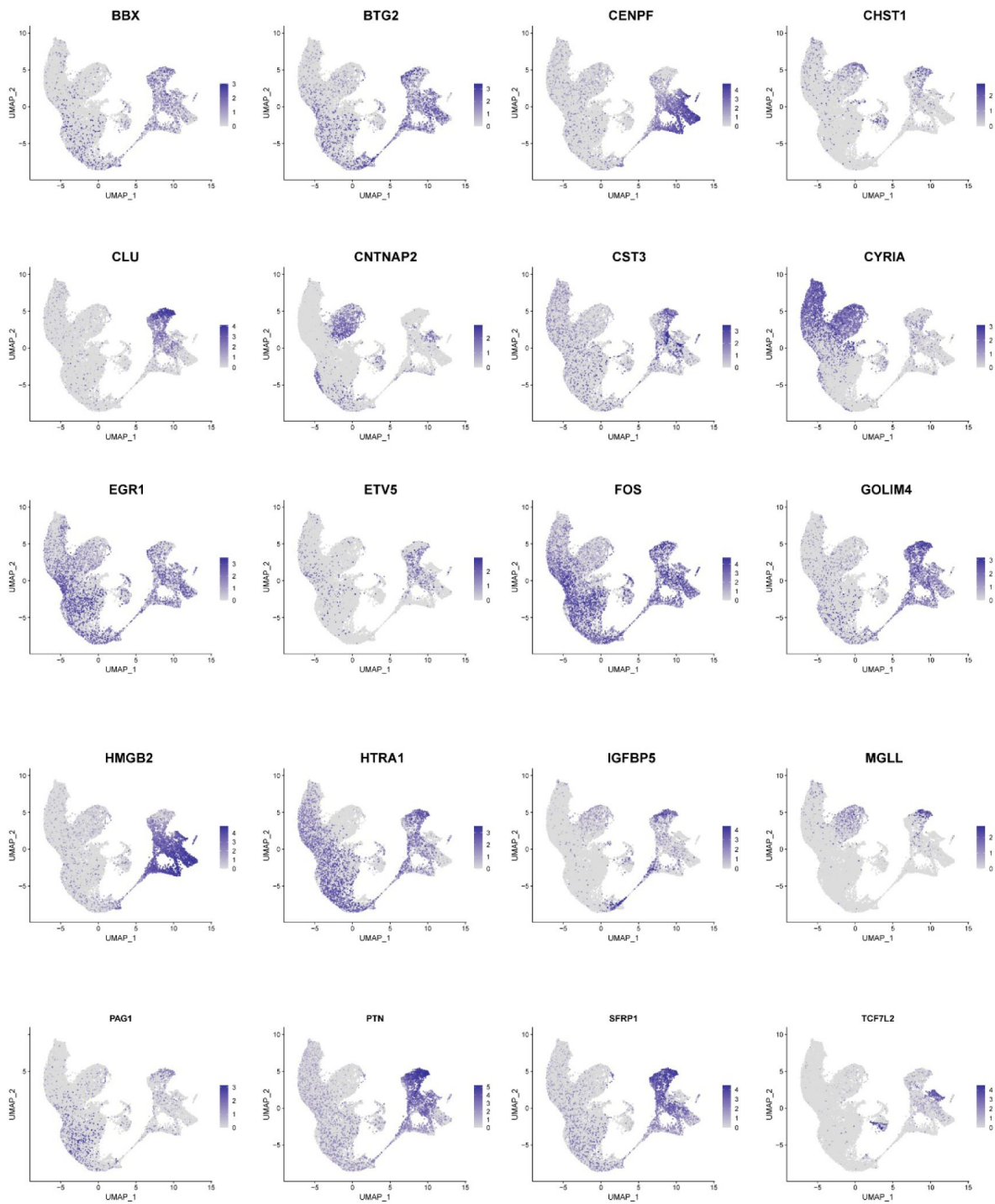


Figure S4.

Stem and Excitatory neuron subclusters mapping genes.

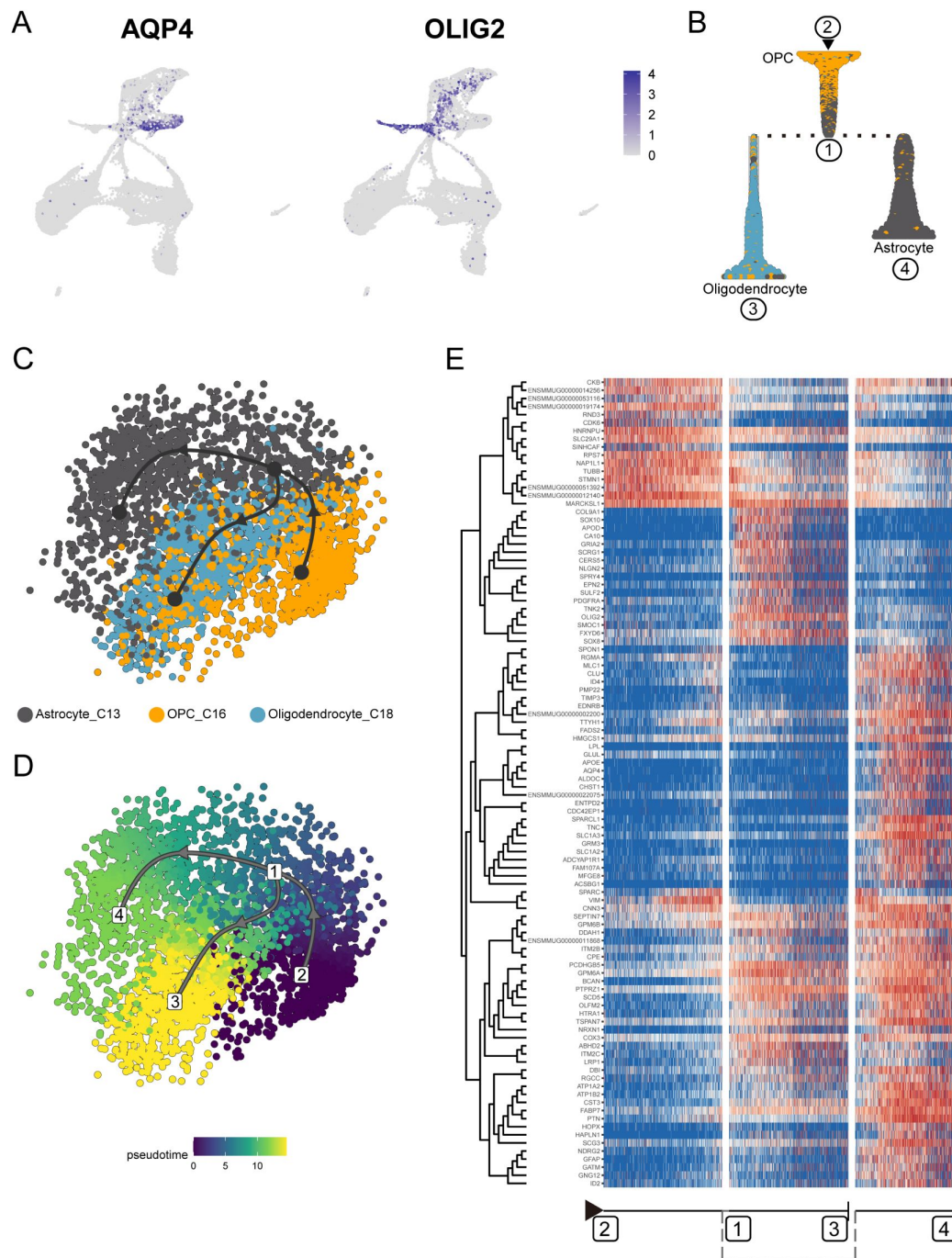


Figure S6.

Transcriptional regulation of OPC differentiation into astrocytes and oligodendrocytes.

(A) Feature plot of classic marker for astrocytes (AQP4) and oligodendrocytes (OLIG2).

(B) Slingshot branching tree related to Slingshot pseudotime analysis in (C) and (D). Root is OPC (C16), tips astrocytes (C13), and oligodendrocytes (C18).

(C and D) Pseudotime analysis by Slingshot of glial cell lineage (C16 → C13, C16 → C12). The Slingshot result indicated the trajectories of lineages, and the arrows indicated directions of the pseudotime. Cells are colored according to their identity (C) and pseudotime (D). Dots: single cells; colors: cluster and subcluster identity.

(E) Heatmap shows the relative expression of top 150 genes displaying significant changes along the pseudotime axis of glial cell lineage (C16 → C13, C16 → C12). The columns represent the cells being ordered along the pseudotime axis.

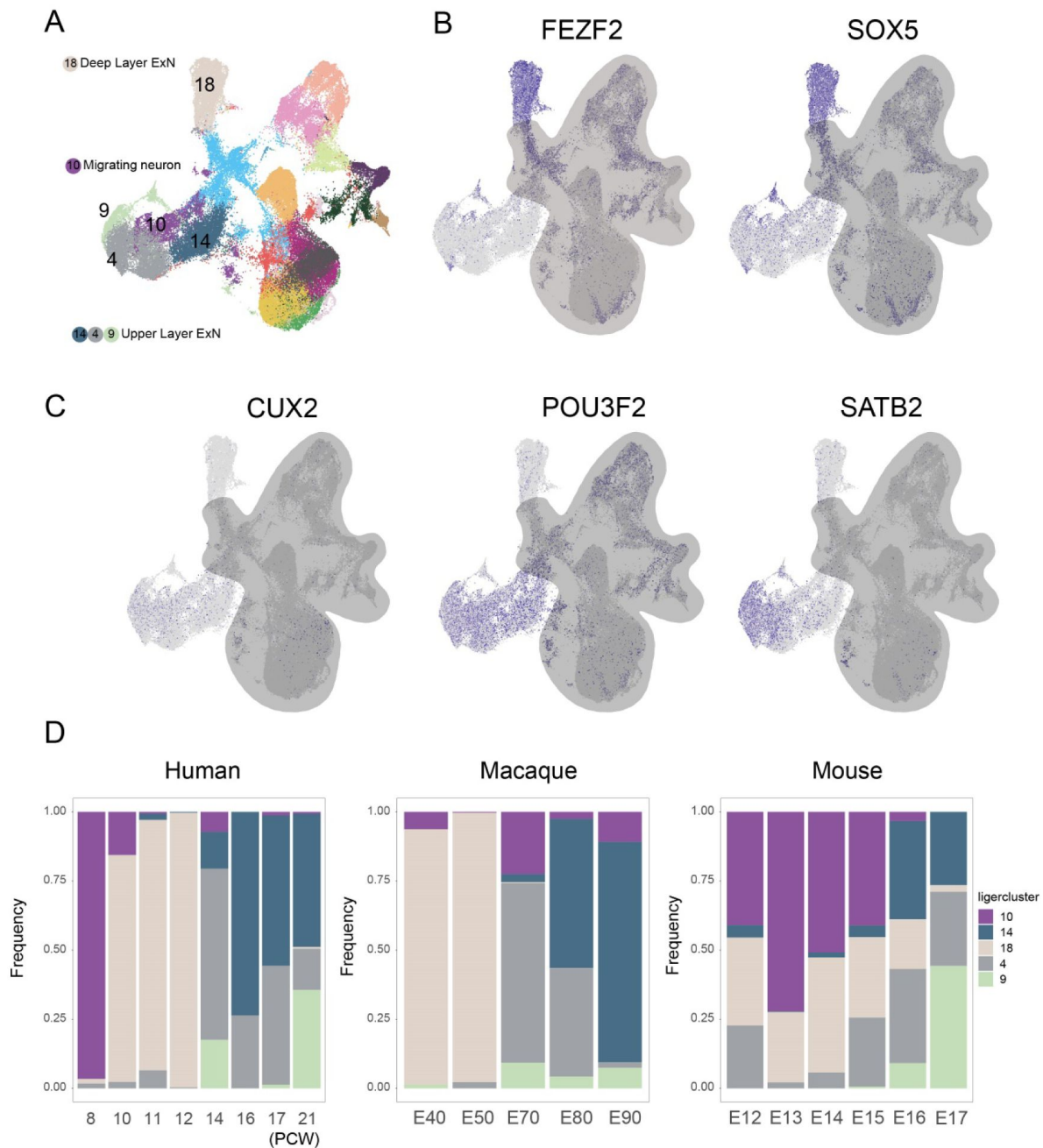


Figure S7.

Upper layer and deep layer excitatory neuron proportion analysis among species

(A) Cell type annotation of the integrated dataset.

(B and C) Featurerplot shows the expression of upper-layer marker genes *CUX2*, *POU3F2*, *SATB2*, and deep-layer marker genes *FEZF2* and *SOX5*. (D) Proportion analysis of excitatory neuron subclusters in human, macaque, and mouse datasets at different developmental time points.

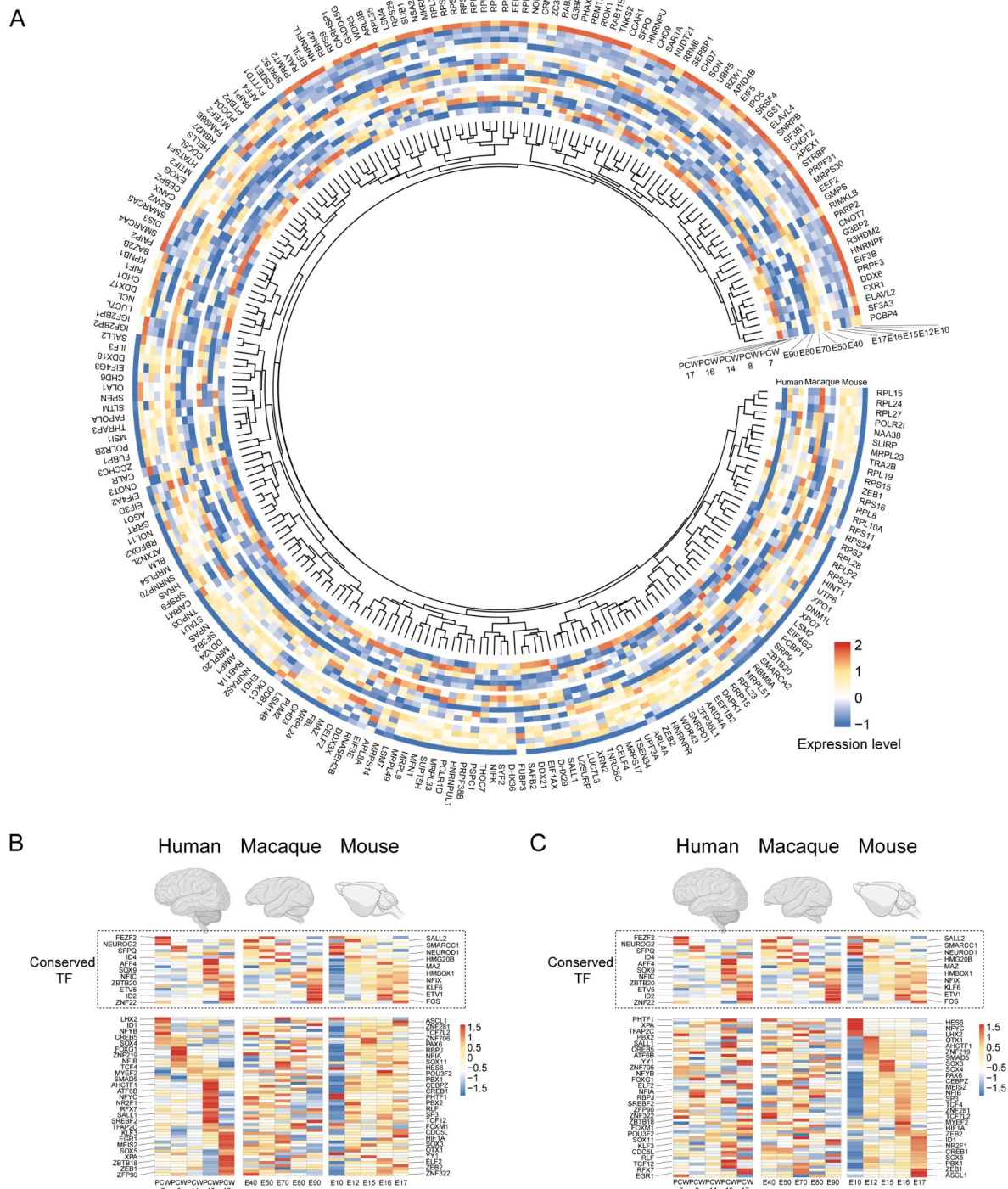


Figure S8.

Temporal expression pattern of RNA binding protein and transcription factor genes in human, macaque, and mouse vRG.

(A) Temporal expression heatmap of RNA binding protein genes in human, macaque, and mouse ventricular radial glia.
 (B and C) Temporal expression heatmap of Transcription factors genes in human, macaque and mouse ventricular radial glia sorted by human (B) and mouse (C) temporal pattern. (Note: each line is the same homologous gene).

Gene expression importance across the vRG to IPC(C10-C22-C8) trajectory related to fig. S5B.

Dataset S5. (separate file)

Gene expression importance across the OPC differentiation into astrocytes and oligodendrocytes lineage trajectory related to fig. S6E.

Dataset S6. (separate file)

The original data of normalized gene expression matrix in human, macaque, and mouse vRG related to **fig.6A** [↗](#).

Dataset S7. (separate file)

Regulon specificity score for each timepoint in human, macaque, and mouse vRG related to **fig. 6** [↗](#) C to H.

References

1. Jabaudon D (2017) **Fate and freedom in developing neocortical circuits** *Nat Commun* **8** <https://doi.org/10.1038/ncomms16042>
2. Rakic P (1974) **Neurons in rhesus monkey visual cortex: systematic relation between time of origin and eventual disposition** *Science* **183**:425–427
3. Telley L. *et al.* (2019) **Temporal patterning of apical progenitors and their daughter neurons in the developing neocortex** *Science* **364** <https://doi.org/10.1126/science.aav2522>
4. Hansen D.V., Lui J.H., Parker P.R.L., Kriegstein A.R (2010) **Neurogenic radial glia in the outer subventricular zone of human neocortex** *Nature* **464**:554–561 <https://doi.org/10.1038/nature08845>
5. Ostrem B.E.L., Lui J.H., Gertz C.C., Kriegstein A.R (2014) **Control of outer radial glial stem cell mitosis in the human brain** *Cell Reports* **8**:656–664 <https://doi.org/10.1016/j.celrep.2014.06.058>
6. Betizeau M. *et al.* (2013) **Precursor diversity and complexity of lineage relationships in the outer subventricular zone of the primate** *Neuron* **80**:442–457 <https://doi.org/10.1016/j.neuron.2013.09.032>
7. Pollen Alex A. *et al.* (2015) **Molecular Identity of Human Outer Radial Glia during Cortical Development** *Cell* **163**:55–67 <https://doi.org/10.1016/j.cell.2015.09.004>
8. Penisson M., Ladewig J., Belvindrah R., Francis F (2019) **Genes and Mechanisms Involved in the Generation and Amplification of Basal Radial Glial Cells** *Front Cell Neurosci* **13** <https://doi.org/10.3389/fncel.2019.00381>
9. Luecken M.D., Theis F.J (2019) **Current best practices in single-cell RNA-seq analysis: a tutorial** *Mol Syst Biol* **15** <https://doi.org/10.15252/msb.20188746>
10. Ruan X., Kang B., Qi C., Lin W., Wang J., Zhang X (2021) **Progenitor cell diversity in the developing mouse neocortex** *Proc Natl Acad Sci U S A* **118** <https://doi.org/10.1073/pnas.2018866118>
11. Di Bella D.J. *et al.* (2021) **Molecular logic of cellular diversification in the mouse cerebral cortex** *Nature* **595**:554–559 <https://doi.org/10.1038/s41586-021-03670-5>
12. Trevino A.E. *et al.* (2021) **Chromatin and gene-regulatory dynamics of the developing human cerebral cortex at single-cell resolution** *Cell* **184** <https://doi.org/10.1016/j.cell.2021.07.039>
13. Nowakowski T.J. *et al.* (2017) **Spatiotemporal gene expression trajectories reveal developmental hierarchies of the human cortex** *Science* **358**:1318–1323 <https://doi.org/10.1126/science.aap8809>

14. Eze U.C., Bhaduri A., Haeussler M., Nowakowski T.J., Kriegstein A.R (2021) **Single-cell atlas of early human brain development highlights heterogeneity of human neuroepithelial cells and early radial glia** *Nature Neuroscience* **24**:584–594 <https://doi.org/10.1038/s41593-020-00794-1>
15. Pebworth M.-P., Ross J., Andrews M., Bhaduri A., Kriegstein A.R (2021) **Human intermediate progenitor diversity during cortical development** *Proceedings of the National Academy of Sciences of the United States of America* **118** <https://doi.org/10.1073/pnas.2019415118>
16. Yuan W. *et al.* (2022) **Temporally divergent regulatory mechanisms govern neuronal diversification and maturation in the mouse and marmoset neocortex** *Nature Neuroscience* **25**:1049–1058 <https://doi.org/10.1038/s41593-022-01123-4>
17. Bakken T.E. *et al.* (2016) **A comprehensive transcriptional map of primate brain development** *Nature* **535**:367–375 <https://doi.org/10.1038/nature18637>
18. Bernard A. *et al.* (2012) **Transcriptional architecture of the primate neocortex** *Neuron* **73**:1083–1099 <https://doi.org/10.1016/j.neuron.2012.03.002>
19. Zeng H. *et al.* (2012) **Large-scale cellular-resolution gene profiling in human neocortex reveals species-specific molecular signatures** *Cell* **149**:483–496 <https://doi.org/10.1016/j.cell.2012.02.052>
20. Hevner R.F (2019) **Intermediate progenitors and Tbr2 in cortical development** *J Anat* **235**:616–625 <https://doi.org/10.1111/joa.12939>
21. Silbereis J.C., Huang E.J., Back S.A., Rowitch D.H (2010) **Towards improved animal models of neonatal white matter injury associated with cerebral palsy** *Disease Models & Mechanisms* **3**:678–688 <https://doi.org/10.1242/dmm.002915>
22. Tsyporin J. *et al.* (2021) **Transcriptional repression by FEZF2 restricts alternative identities of cortical projection neurons** *Cell Rep* **35** <https://doi.org/10.1016/j.celrep.2021.109269>
23. Li Z., Tyler W.A., Zeldich E., Santpere Baró G., Okamoto M., Gao T., Li M., Sestan N., Haydar T.F (2020) **Transcriptional priming as a conserved mechanism of lineage diversification in the developing mouse and human neocortex** *Sci Adv* **6** <https://doi.org/10.1126/sciadv.abd2068>
24. He Z. *et al.* (2017) **Comprehensive transcriptome analysis of neocortical layers in humans, chimpanzees and macaques** *Nat Neurosci* **20**:886–895 <https://doi.org/10.1038/nn.4548>
25. Liu J. *et al.* (2017) **The Primate-Specific Gene TMEM14B Marks Outer Radial Glia Cells and Promotes Cortical Expansion and Folding** *Cell Stem Cell* **21**:635–649 <https://doi.org/10.1016/j.stem.2017.08.013>
26. Johnson M.B. *et al.* (2018) **Aspm knockout ferret reveals an evolutionary mechanism governing cerebral cortical size** *Nature* **556**:370–375 <https://doi.org/10.1038/s41586-018-0035-0>
27. Guo Z. *et al.* (2021) **RGCC balances self-renewal and neuronal differentiation of neural stem cells in the developing mammalian neocortex** *EMBO Rep* **22** <https://doi.org/10.15252/embr.202051781>

28. Zhou X. *et al.* (2020) **Cellular and molecular properties of neural progenitors in the developing mammalian hypothalamus** *Nat Commun* **11** <https://doi.org/10.1038/s41467-020-17890-2>
29. Weng Q. *et al.* (2019) **Single-Cell Transcriptomics Uncovers Glial Progenitor Diversity and Cell Fate Determinants during Development and Gliomagenesis** *Cell Stem Cell* **24**:707–723 <https://doi.org/10.1016/j.stem.2019.03.006>
30. Gadea A., Schinelli S., Gallo V (2008) **Endothelin-1 regulates astrocyte proliferation and reactive gliosis via a JNK/c-Jun signaling pathway** *J Neurosci* **28**:2394–2408 <https://doi.org/10.1523/JNEUROSCI.5652-07.2008>
31. Telley L. *et al.* (2019) **Temporal patterning of apical progenitors and their daughter neurons in the developing neocortex** *Science* **364** <https://doi.org/10.1126/science.aav2522>
32. Lin Y., Yang J., Shen Z., Ma J., Simons B.D., Shi S.H (2021) **Behavior and lineage progression of neural progenitors in the mammalian cortex** *Curr Opin Neurobiol* **66**:144–157 <https://doi.org/10.1016/j.conb.2020.10.017>
33. Shu P. *et al.* (2019) **Opposing Gradients of MicroRNA Expression Temporally Pattern Layer Formation in the Developing Neocortex** *Dev Cell* **49**:764–785 <https://doi.org/10.1016/j.devcel.2019.04.017>
34. Welch J.D., Kozareva V., Ferreira A., Vanderburg C., Martin C., Macosko E.Z (2019) **Single-Cell Multi-omic Integration Compares and Contrasts Features of Brain Cell Identity** *Cell* **177** <https://doi.org/10.1016/j.cell.2019.05.006>
35. Zhong S. *et al.* (2018) **A single-cell RNA-seq survey of the developmental landscape of the human prefrontal cortex** *Nature* **555**:524–528 <https://doi.org/10.1038/nature25980>
36. Fan X. *et al.* (2020) **Single-cell transcriptome analysis reveals cell lineage specification in temporal-spatial patterns in human cortical development** *Sci Adv* **6** <https://doi.org/10.1126/sciadv.aaz2978>
37. Wang X., Tsai J.-W., LaMonica B., Kriegstein A.R (2011) **A new subtype of progenitor cell in the mouse embryonic neocortex** *Nature Neuroscience* **14**:555–561 <https://doi.org/10.1038/nn.2807>
38. Zhu Y. *et al.* (2018) **Spatiotemporal transcriptomic divergence across human and macaque brain development** *Science* **362** <https://doi.org/10.1126/science.aat8077>
39. Luo X. *et al.* (2021) **3D Genome of macaque fetal brain reveals evolutionary innovations during primate corticogenesis** *Cell* **184** <https://doi.org/10.1016/j.cell.2021.01.001>
40. Bhaduri A., Sandoval-Espinosa C., Otero-Garcia M., Oh I., Yin R., Eze U.C., Nowakowski T.J., Kriegstein A.R (2021) **An atlas of cortical arealization identifies dynamic molecular signatures** *Nature* **598**:200–204 <https://doi.org/10.1038/s41586-021-03910-8>
41. Hansen D.V., Lui J.H., Parker P.R., Kriegstein A.R (2010) **Neurogenic radial glia in the outer subventricular zone of human neocortex** *Nature* **464**:554–561 <https://doi.org/10.1038/nature08845>

42. Nowakowski T.J., Pollen A.A., Sandoval-Espinosa C., Kriegstein A.R (2016) **Transformation of the Radial Glia Scaffold Demarcates Two Stages of Human Cerebral Cortex Development** *Neuron* **91**:1219–1227 <https://doi.org/10.1016/j.neuron.2016.09.005>
43. Wang X., Tsai J.W., LaMonica B., Kriegstein A.R (2011) **A new subtype of progenitor cell in the mouse embryonic neocortex** *Nat Neurosci* **14**:555–561 <https://doi.org/10.1038/nn.2807>
44. Kalebic N., Gilardi C., Albert M., Namba T., Long K.R., Kostic M., Langen B., Huttner W.B (2018) **Human-specific ARHGAP11B induces hallmarks of neocortical expansion in developing ferret neocortex** *Elife* **7** <https://doi.org/10.7554/eLife.41241>
45. Xing L., Kubik-Zahorodna A., Namba T., Pinson A., Florio M., Prochazka J., Sarov M., Sedlacek R., Huttner W.B (2021) **Expression of human-specific ARHGAP11B in mice leads to neocortex expansion and increased memory flexibility** *EMBO J* **40** <https://doi.org/10.15252/embj.2020107093>
46. Heide M., Haffner C., Murayama A., Kurotaki Y., Shinohara H., Okano H., Sasaki E., Huttner W.B (2020) **Human-specific increases size and folding of primate neocortex in the fetal marmoset** *Science* **369**:546–550 <https://doi.org/10.1126/science.abb2401>
47. Vitali I. *et al.* (2018) **Progenitor Hyperpolarization Regulates the Sequential Generation of Neuronal Subtypes in the Developing Neocortex** *Cell* **174**:1264–1276 <https://doi.org/10.1016/j.cell.2018.06.036>
48. Molyneaux B.J., Arlotta P., Hirata T., Hibi M., Macklis J.D (2005) **Fezl is required for the birth and specification of corticospinal motor neurons** *Neuron* **47**:817–831 <https://doi.org/10.1016/j.neuron.2005.08.030>
49. Leone D.P., Srinivasan K., Chen B., Alcamo E., McConnell S.K (2008) **The determination of projection neuron identity in the developing cerebral cortex** *Curr Opin Neurobiol* **18**:28–35 <https://doi.org/10.1016/j.conb.2008.05.006>
50. Bakken T.E. *et al.* (2016) **A comprehensive transcriptional map of primate brain development** *Nature* **535**:367–375 <https://doi.org/10.1038/nature18637>
51. Lein E.S., Belgard T.G., Hawrylycz M., Molnar Z (2017) **Transcriptomic Perspectives on Neocortical Structure, Development, Evolution, and Disease** *Annu Rev Neurosci* **40**:629–652 <https://doi.org/10.1146/annurev-neuro-070815-013858>
52. Satija R., Farrell J.A., Gennert D., Schier A.F., Regev A (2015) **Spatial reconstruction of single-cell gene expression data** *Nat Biotechnol* **33**:495–502 <https://doi.org/10.1038/nbt.3192>
53. Street K., Risso D., Fletcher R.B., Das D., Ngai J., Yosef N., Purdom E., Dudoit S (2018) **Slingshot: cell lineage and pseudotime inference for single-cell transcriptomics** *BMC Genomics* **19** <https://doi.org/10.1186/s12864-018-4772-0>
54. Saelens W., Cannoodt R., Todorov H., Saeys Y (2019) **A comparison of single-cell trajectory inference methods** *Nat Biotechnol* **37**:547–554 <https://doi.org/10.1038/s41587-019-0071-9>
55. Verfaillie A., Imrichova H., Janky R., Aerts S (2015) **iRegulon and i-cisTarget: Reconstructing Regulatory Networks Using Motif and Track Enrichment** *Curr Protoc Bioinformatics* **52**:2–12 <https://doi.org/10.1002/0471250953.bi0216s52>

- 56. Shannon P., Markiel A., Ozier O., Baliga N.S., Wang J.T., Ramage D., Amin N., Schwikowski B., Ideker T (2003) **Cytoscape: a software environment for integrated models of biomolecular interaction networks** *Genome Res* **13**:2498–2504 <https://doi.org/10.1101/gr.1239303>
- 57. Kumar L., E Futschik M. (2007) **Mfuzz: a software package for soft clustering of microarray data** *Bioinformatics* **2**:5–7

Article and author information

Longjiang Xu

Institute of Medical Biology Chinese Academy of Medical Sciences, Chinese Academy of Medical Science and Peking Union Medical College, Kunming, China
ORCID iD: [0000-0002-3534-6840](https://orcid.org/0000-0002-3534-6840)

Zan Yuan

Department of Molecular Biology and Biochemistry, Institute of Basic Medical Sciences, Medical Primate Research Center, Neuroscience Center, Chinese Academy of Medical Sciences, School of Basic Medicine Peking Union Medical College, Beijing, China, Agricultural Bioinformatics Key Laboratory of Hubei Province, Hubei Engineering Technology Research Center of Agricultural Big Data, College of Informatics, Huazhong Agricultural University, Wuhan, China
ORCID iD: [0000-0001-5582-5120](https://orcid.org/0000-0001-5582-5120)

Jiafeng Zhou

Department of Molecular Biology and Biochemistry, Institute of Basic Medical Sciences, Medical Primate Research Center, Neuroscience Center, Chinese Academy of Medical Sciences, School of Basic Medicine Peking Union Medical College, Beijing, China, State Key Laboratory of Common Mechanism Research for Major Diseases, Chinese Academy of Medical Sciences, Peking Union Medical College, Beijing, China

Yuan Zhao

Institute of Medical Biology Chinese Academy of Medical Sciences, Chinese Academy of Medical Science and Peking Union Medical College, Kunming, China

Wei Liu

Department of Molecular Biology and Biochemistry, Institute of Basic Medical Sciences, Medical Primate Research Center, Neuroscience Center, Chinese Academy of Medical Sciences, School of Basic Medicine Peking Union Medical College, Beijing, China, State Key Laboratory of Common Mechanism Research for Major Diseases, Chinese Academy of Medical Sciences, Peking Union Medical College, Beijing, China

Shuaiyao Lu

Institute of Medical Biology Chinese Academy of Medical Sciences, Chinese Academy of Medical Science and Peking Union Medical College, Kunming, China

Zhanlong He

Institute of Medical Biology Chinese Academy of Medical Sciences, Chinese Academy of Medical Science and Peking Union Medical College, Kunming, China

Boqin Qiang

Department of Molecular Biology and Biochemistry, Institute of Basic Medical Sciences, Medical Primate Research Center, Neuroscience Center, Chinese Academy of Medical Sciences, School of Basic Medicine Peking Union Medical College, Beijing, China, State Key Laboratory of Common Mechanism Research for Major Diseases, Chinese Academy of Medical Sciences, Peking Union Medical College, Beijing, China

Pengcheng Shu

Department of Molecular Biology and Biochemistry, Institute of Basic Medical Sciences, Medical Primate Research Center, Neuroscience Center, Chinese Academy of Medical Sciences, School of Basic Medicine Peking Union Medical College, Beijing, China, State Key Laboratory of Common Mechanism Research for Major Diseases, Chinese Academy of Medical Sciences, Peking Union Medical College, Beijing, China, Chinese Institute for Brain Research, Beijing, China

For correspondence: pengcheng_shu@ibms.pumc.edu.cn

ORCID iD: [0000-0002-3091-2559](https://orcid.org/0000-0002-3091-2559)

Yang Chen

Department of Molecular Biology and Biochemistry, Institute of Basic Medical Sciences, Medical Primate Research Center, Neuroscience Center, Chinese Academy of Medical Sciences, School of Basic Medicine Peking Union Medical College, Beijing, China, State Key Laboratory of Common Mechanism Research for Major Diseases, Chinese Academy of Medical Sciences, Peking Union Medical College, Beijing, China

For correspondence: pengcheng_shu@ibms.pumc.edu.cn

Xiaozhong Peng

Institute of Medical Biology Chinese Academy of Medical Sciences, Chinese Academy of Medical Science and Peking Union Medical College, Kunming, China, Department of Molecular Biology and Biochemistry, Institute of Basic Medical Sciences, Medical Primate Research Center, Neuroscience Center, Chinese Academy of Medical Sciences, School of Basic Medicine Peking Union Medical College, Beijing, China, State Key Laboratory of Respiratory Health and Multimorbidity, Chinese Academy of Medical Sciences, Peking Union Medical College, Beijing, China, Institute of Laboratory Animal Science, Chinese Academy of Medical Sciences & Peking Union Medical College, Beijing, China

For correspondence: pengxiaozhong@pumc.edu.cn

Copyright

© 2023, Xu et al.

This article is distributed under the terms of the [Creative Commons Attribution License](https://creativecommons.org/licenses/by/4.0/), which permits unrestricted use and redistribution provided that the original author and source are credited.

Editors

Reviewing Editor

Sacha Nelson

Brandeis University, Waltham, United States of America

Senior Editor

Sacha Nelson

Brandeis University, Waltham, United States of America

Reviewer #1 (Public Review):

In the article "Temporal transcriptomic dynamics in developing macaque neocortex", Xu et al. analyze the cellular composition and transcriptomic profiles of the developing macaque parietal cortex using single-cell RNA sequencing. The authors profiled eight prenatal rhesus macaque brains at five timepoints (E40, E50, E70, E80, and E90) and obtained a total of around 53,000 high-quality cells for downstream analysis. The dataset provides a high-resolution view into the developmental processes of early and mid-fetal macaque cortical development and will potentially be a valuable resource for future comparative studies of primate neurogenesis and neural stem cell fate specification. Their analysis of this dataset focused on the temporal gene expression profiles of outer and ventricular radial glia and utilized pseudotime trajectory analysis to characterize the genes associated with radial glial and neuronal differentiation. The rhesus macaque dataset presented in this study was then integrated with prenatal mouse and human scRNA-seq datasets to probe species differences in ventricular radial glia to intermediate progenitor cell trajectories. Additionally, the expression profile of macaque radial glia across time was compared to those of mouse apical progenitors to identify conserved and divergent expression patterns of transcription factors.

The main findings of this paper corroborate many previously reported and fundamental features of primate neurogenesis: deep layer neurons are generated before upper layer excitatory neurons, the expansion of outer radial glia in the primate lineage, conserved molecular markers of outer radial glia, and the early specification of progenitors. Furthermore, the authors show some interesting divergent features of macaque radial glial gene regulatory networks as compared to mouse. Overall, despite some uncertainties surrounding the clustering and annotations of certain cell types, the manuscript provides a valuable scRNA-seq dataset of early prenatal rhesus macaque brain development. The dynamic expression patterns and trajectory analysis of ventricular and outer radial glia provide valuable data and lists of differentially expressed genes (some consistent with previous studies, others reported for the first time here) for future studies.

<https://doi.org/10.7554/eLife.90325.2.sa2>

Reviewer #2 (Public Review):

Summary:

This manuscript by Xu et al., is an interesting study aiming to identify novel features of macaque cortical development. This study serves as a valuable atlas of single cell data during macaque neurogenesis, which extends the developmental stages previously explored. Overall, the authors have achieved their aim of collecting a comprehensive dataset of macaque cortical neurogenesis and have identified a few unknown features of macaque development.

Strengths:

The authors have accumulated a robust dataset of developmental time points and have applied a variety of informatic approaches to interrogate this dataset. One interesting finding in this study is the expression of previously unknown receptors on macaque oRG cells. Another novel aspect of this paper is the temporal dissection of neocortical development across species. The identification that the regulome looks quite different, despite similar expression of transcription factors in discrete cell types, is intriguing.

Reviewer #3 (Public Review):

Summary:

The study adds to the existing data that have established that cortical development in rhesus macaque is known to recapitulate multiple facets cortical development in humans. The authors generate and analyze single cell transcriptomic data from the timecourse of embryonic neurogenesis.

Strengths:

Studies of primate developmental biology are hindered by the limited availability and limit replication. In this regard, a new dataset is useful.

The study analyzes parietal cortex, while previous studies focused on frontal and motor cortex. This may be the first analysis of macaque parietal cortex and, as such, may provide important insights into arealization, which the authors have not addressed

<https://doi.org/10.7554/eLife.90325.2.sa0>

Author Response

Thanks to all the reviewers for their insightful and constructive comments, which are very helpful in improving the manuscript. We are encouraged by the many positive comments regarding the significance of our findings and the value of our data. Regarding the reviews' concern on cell classification, we used several additional marker genes to explain the identification of cell clusters and subclusters. We have further analyzed and rewrote part of the text to address the concerns raised. Here is a point-by-point response to the reviewers' comments and concerns. Figures R1-R9 were provided only for additional information for reviewers and were not included in the revised manuscript.

Reviewer #1 (Public Review):

In the article "Temporal transcriptomic dynamics in developing macaque neocortex", Xu et al. analyze the cellular composition and transcriptomic profiles of the developing macaque parietal cortex using single-cell RNA sequencing. The authors profiled eight prenatal rhesus macaque brains at five timepoints (E40, E50, E70, E80, and E90) and obtained a total of around 53,000 high-quality cells for downstream analysis. The dataset provides a high-resolution view into the developmental processes of early and mid-fetal macaque cortical development and will potentially be a valuable resource for future comparative studies of primate neurogenesis and neural stem cell fate specification. Their analysis of this dataset focused on the temporal gene expression profiles of outer and ventricular radial glia and utilized pseudotime trajectory analysis to characterize the genes associated with radial glial and neuronal differentiation. The rhesus macaque dataset presented in this study was then integrated with prenatal mouse and human scRNA-seq datasets to probe species differences in ventricular radial glia to intermediate progenitor cell trajectories. Additionally, the expression profile of macaque radial glia across time was compared to those of mouse apical progenitors to identify conserved and divergent expression patterns of transcription factors.

The main findings of this paper corroborate many previously reported and fundamental features of primate neurogenesis: deep layer neurons are generated before upper layer excitatory neurons, the expansion of outer radial glia in the primate lineage, conserved

molecular markers of outer radial glia, and the early specification of progenitors. Furthermore, the authors show some interesting divergent features of macaque radial glial gene regulatory networks as compared to mouse. Overall, despite some uncertainties surrounding the clustering and annotations of certain cell types, the manuscript provides a valuable scRNA-seq dataset of early prenatal rhesus macaque brain development. The dynamic expression patterns and trajectory analysis of ventricular and outer radial glia provide valuable data and lists of differentially expressed genes (some consistent with previous studies, others reported for the first time here) for future studies.

The major weaknesses of this study are the inconsistent dissection of the targeted brain region and the loss of more mature excitatory neurons in samples from later developmental timepoint due to the use of single-cell RNA-seq. The authors mention that they could observe ventral progenitors and even midbrain neurons in their analyses. Ventral progenitors should not be present if the authors had properly dissected the parietal cortex. The fact that they obtained even midbrain cells point to an inadequate dissection or poor cell classification. If this is the result of poor classification, it could be easily fixed by using more markers with higher specificity. However, if it is the result of a poor dissection, some of the cells in other clusters could potentially be from midbrain as well. The loss of more mature excitatory neurons is also problematic because on top of hindering the analysis of these neurons in later developmental periods, it also affects the cell proportions the authors use to support some of their claims. The study could also benefit from the validation of some of the genes the authors uncovered to be specifically expressed in different populations of radial glia.

We thank the Reviewer's comments and apologize for the shortcomings of tissue dissection and cell capture.

We used more marker genes for major cell classification, such as SHOX2, IGFBP5, TAC1, PNYN, FLT1, and CYP1B, in new Figure 1D, to improve the cell type annotation results. We improved the cell type annotation results by fixing cluster 20 from C20 as Ventral LGE-derived interneuron precursor and cluster by the expression of IGFBP5, TAC1, and PDYN; fixing cluster 23 from meningeal cells to thalamus cells by the expression of ZIC2, ZIC4, and SHOX2. These cell types were excluded in the follow-up analysis. Due to EN8 being previously incorrectly defined as midbrain neurons, it resulted in a misunderstanding of the dissection result as a poor dissection. After carefully reviewing the data analysis process, we determined that EN8 was a small group of cells in cluster 23 mistakenly selected during excitatory neuron analysis, as shown in Figure R5(A), which was corrected after revision. In the revised manuscript, we deleted the previous EN8 subcluster and renumbered the rest of the excitatory neuron subclusters in the new Figure 2.

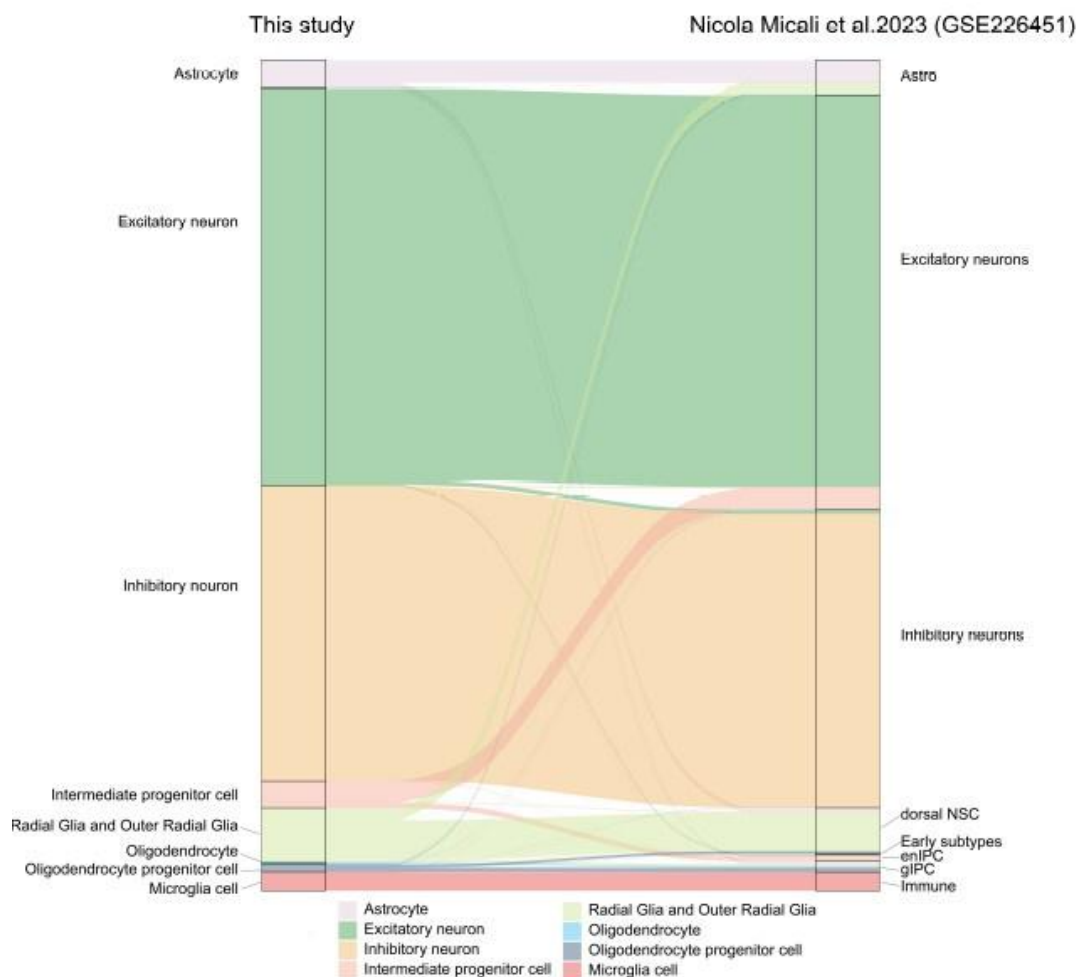
In addition, we also improved the description of sample collection as follows: "We collected eight pregnancy-derived fetal brains of rhesus macaque (*Macaca mulatta*) at five prenatal developmental stages (E40, E50, E70, E80, E90) and dissected the parietal lobe cortex. Because of the different development times of rhesus monkeys, prenatal cortex size and morphology are different. To ensure that the anatomical sites of each sample are roughly the same, we use the lateral groove as a reference to collect the parietal lobe for single-cell sequencing (as indicated by bright yellow in Figure S1A) and do not make a clear distinction between the different regional parts including primary somatosensory cortex and association cortices in the process of sampling". As shown in Figure S1A, due to the small volume of the cerebral cortex at early time points, especially in E40, a small number of cells beyond the dorsal parietal lobe, including the ventral cortex cells and thalamus cells, were collected during the sampling process with the brain stereotaxic instrument.

In this study, the BD method was used to capture single cells. Due to the fixed size of the micropores, this method might be less efficient in capturing mature excitatory neurons. However, it has a good capture effect on newborn neurons at each sampling time point so that the generation of excitatory neurons at different developmental time points can be well observed, as shown in Figure 2, which aligns with our research purpose.

To verify the reliability of our cell annotation results, we compared the similarity of cell-type association between our study and recently published research(Micali N, Ma S, Li M, et al. Science. doi:10.1126/science.adf3786.PMID: 37824652), using the scmap package to project major cell types in our macaque development scRNA-seq dataset to GSE226451. The river plot in Author response image 1 illustrates the broadly similar relationships of cell type classification between the two datasets.

Author response image 1.

Riverplot illustrates relationships between datasets in this study and recently published developing macaque telencephalon datasets major cell type annotation.

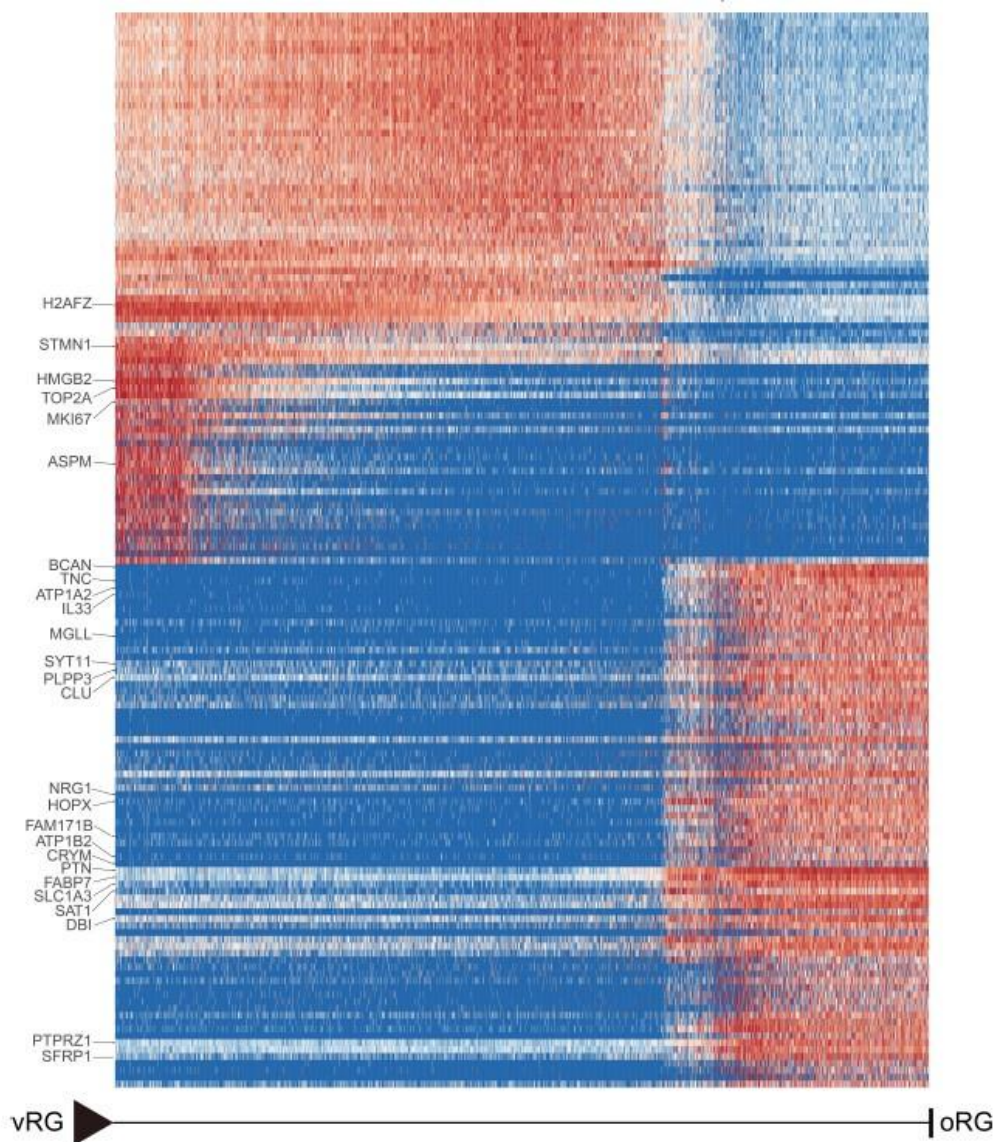


Furthermore, bioinformatics analysis is used for the validation of genes specifically expressed in outer radial glia. We verified terminal oRG differentiation genes in the recently published macaque telencephalic development dataset(Micali N, Ma S, Li M, et al. Science. doi:10.1126/science.adf3786.PMID: 37824652) (GEO accession: GSE226451). The results of

Author response image 2 show that the gene expression showed states/stages. Most of the oRG terminal differentiation markers genes identified in our study were also expressed in the oRG cells of the GSE226451 dataset. In particular, the two datasets were consistent in the expression of ion channel genes ATP1A2, ATP1A2, and SCN4B.

Author response image 2.

Heatmap shows the relative expression of genes displaying significant changes along the pseudotime axis of vRG to oRG from the dataset of Nicola Micali et al.2023(GEO: GSE226451). The columns represent the cells being ordered along the pseudotime axis.



Reviewer #2 (Public Review):

Summary:

This manuscript by Xu et al., is an interesting study aiming to identify novel features of macaque cortical development. This study serves as a valuable atlas of single cell data during macaque neurogenesis, which extends the developmental stages previously explored. Overall, the authors have achieved their aim of collecting a comprehensive

dataset of macaque cortical neurogenesis and have identified a few unknown features of macaque development.

Strengths:

The authors have accumulated a robust dataset of developmental time points and have applied a variety of informatic approaches to interrogate this dataset. One interesting finding in this study is the expression of previously unknown receptors on macaque oRG cells. Another novel aspect of this paper is the temporal dissection of neocortical development across species. The identification that the regulome looks quite different, despite similar expression of transcription factors in discrete cell types, is intriguing.

Weaknesses:

Due to the focus on demonstrating the robustness of the dataset, the novel findings in this manuscript are underdeveloped. There is also a lack of experimental validation. This is a particular weakness for newly identified features (like receptors in oRG cells). It's important to show expression in relevant cell types and, if possible, perform functional perturbations on these cell types. The presentation of the data highlighting novel findings could also be clarified at higher resolution, and dissected through additional informatic analyses. Additionally, the presentation of ideas and goals of this manuscript should be further clarified. A major gap in the study rationale and results is that the data was collected exclusively in the parietal lobe, yet the rationale and interpretation of what this data indicates about this specific cortical area was not discussed. Last, a few textual errors about neural development are also present and need to be corrected.

We thank you for your comments and suggestions concerning our manuscript. The comments and suggestions are all valuable and helpful for revising and improving our paper and the essential guiding significance to our research. We have studied the comments carefully and made corrections, which we hope to meet with approval. We have endeavored to address the multiple points raised by the referee.

To support the reliability of our data and newly identified features, we verified terminal oRG differentiation genes in the recently published macaque telencephalic development dataset (Micali N, Ma S, Li M, et al. Science. doi:10.1126/science.adf3786.PMID: 37824652) (GEO accession: GSE226451). The results of Figure R2 show that the gene expression showed states/stages. Most of the oRG terminal differentiation markers genes identified in our study were also expressed in the oRG cells of the GSE226451 dataset. In particular, the two datasets were consistent in the expression of ion channel genes ATP1A2, ATP1A2, and SCN4B.

Our research results mainly explore the conserved features of neocortex development across species. By comparing evolution, we found the types of neural stem cells in the intermediate state, their generative trajectories, and gene expression dynamics accompanying cell trajectories. We further explored the stages of transcriptional dynamics during vRG generating oRG. More analysis was performed through transcriptional factor regulatory network analysis. We performed the TFs regulation network analysis of human vRG with pycenic workflow. The top transcription factors of every time point in human vRG were calculated, and we used the top 10 TFs and their top 5 target genes to perform interaction analysis and generate the regulation network of human vRG in revised figure 6. In comparison of the pycenic results of mouse, macaque and human vRG, it was obvious that the regulatory networks were not evolutionarily conservative. Compared with macaque, the regulatory network of transcription factors and target genes in humans is more complex. Some conserved regulatory relationships present in more than one species are identified, such as HMGN3, EMX2, SOX2, and HMGA2 network at an early stage when deep layer generation and SOX10, ZNF672, ZNF672 network at a late stage when upper-layer generation.

Although the parietal lobe is the center of the somatic senses and is significant for interpreting words as well as language understanding and processing. In this study, the parietal lobe area was selected mainly because of the convenience of sampling the dorsal neocortex. As we described in the Materials and Methods section as follows: “Because of the different development times of rhesus monkeys, prenatal cortex size and morphology are different. To ensure that the anatomical sites of each sample are roughly the same, we use the lateral groove as a reference to collect the parietal lobe for single-cell sequencing (as indicated by bright yellow in Figure S1A) and do not make a clear distinction between the different regional parts including primary somatosensory cortex and association cortices in the process of sampling”.

Thanks for carefully pointing out our manuscript's textual errors about neural development. We have corrected them which were described in the following response.

Reviewer #3 (Public Review):

Summary: The study adds to the existing data that have established that cortical development in rhesus macaque is known to recapitulate multiple facets cortical development in humans. The authors generate and analyze single cell transcriptomic data from the timecourse of embryonic neurogenesis.

Strengths:

Studies of primate developmental biology are hindered by the limited availability and limit replication. In this regard, a new dataset is useful.

The study analyzes parietal cortex, while previous studies focused on frontal and motor cortex. This may be the first analysis of macaque parietal cortex and, as such, may provide important insights into arealization, which the authors have not addressed.

Weaknesses:

The number of cells in the analysis is lower than recent published studies which may limit cell representation and potentially the discovery of subtle changes.

The macaque parietal cortex data is compared to human and mouse pre-frontal cortex. See data from PMCID: PMC8494648 that provides a better comparison.

A deeper assessment of these data in the context of existing studies would help others appreciate the significance of the work.

We thank the reviewer for these suggestions and constructive comments. We agree with the reviewer that the cell number in our study is lower than in recently published studies. The scRNA sequencing in this study was completed between 2018 and 2019, the early stages of the single-cell sequencing technology application. Besides, we have been unable to get extra macaque embryos to enlarge the sample numbers recently since rhesus monkey samples are scarce. Therefore, the number of cells in our study is relatively small compared to recently published single-cell studies.

The dataset suggested by the reviewers is extremely valuable, and we tried to perform analysis as the reviewer suggested to explore temporal expression patterns in different species of parietal cortex. The dataset from PMCID: PMC8494648 provides the developing human brain across regions from gestation week (GW)14 to gestation week (GW)25. Since this data set only covers the middle and late stages of embryonic neurogenesis, it did not fully match the developmental time points of our study for integration analysis. However, we quoted the results of this study in the discussion section.

The human regulation analysis with pyscenic workflow was added into new figure 6 for the comparison of different species vRG regulatory network. Compared with macaque, the regulatory network of transcription factors and target genes in humans is more complex. Some conserved regulatory relationships present in more than one species are identified, such as HMGN3, EMX2, SOX2, and HMGA2 network at an early stage when deep layer generation and SOX10, ZNF672, ZNF672 network at a late stage when upper-layer generation.

Besides, we performed additional integration analysis of our dataset with the recently published macaque neocortex development dataset (GEO accession: GSE226451) to verify the reliability of our cell annotation results and terminal oRG differentiation genes. The river plot in Figure R1 illustrates the broadly similar relationships of cell type classification between the two datasets. The result in Figure R2 showed that most of the oRG terminal differentiation markers genes identified in our study were also expressed in the oRG cells of the GSE226451 dataset. In particular, the two datasets were consistent in the expression of ion channel genes ATP1A2, ATP1A2, and SCN4B.

Reviewer #1 (Recommendations For The Authors):

1. Throughout the manuscript, the term "embryonic" or "embryogenesis" is used in reference to all timepoints (E40-E90) in this study. The embryonic period is a morphologically and anatomically defined developmental period that ends ~E48-E50 in rhesus macaque. Prenatal or developing is a more accurate term when discussing all timepoints of this study.

We thank the reviewer for pointing out this terminology that needs to be clarified. We have now replaced "embryonic" with "prenatal" as a more appropriate description for the sampling time points in the manuscript.

1. *Drosophila* should be italicized in the introduction.

Thanks for suggesting that we have set the "Drosophila" words to italics in the manuscript.

1. Introduction - "In rodents, radial glia are found in the ventricular zone (VZ), where they undergo proliferation and differentiation." This sentence implies that only within rodents are radial glia found within the ventricular zone. Radial glia are present within the ventricular zone of all mammals.

Thanks for careful reading. This sentence has been corrected "In mammals, radial glial cells are found in the ventricular zone (VZ), where they undergo proliferation and differentiation."

1. Figure 1A - an image of the E40 brain is missing.

We first sampled the prenatal developmental cortex of rhesus monkeys at the E40 timepoint. Unfortunately, we forgot to save the photo of the sampling at the E40 time point.

1. Figure 1B and 1C - it is unclear why cluster 20 is not annotated in Figure 1 as in the text it is stated "Each of the 28 identified clusters could be assigned to a cell type identity..." This cluster expresses VIM and PAX6 suggestive of ventricular radial glia and is located topographically approximate to IPC cluster 8 and seems to bridge the gap between neural stem cells and the interneuron clusters. Additionally, cluster 20 appears to be subclustered by itself in the progenitor subcluster UMAP (Figure 3A) suggestive of a batch effect or cells with low quality. The investigation, quality control, and proper annotation of this cluster 20 is necessary.

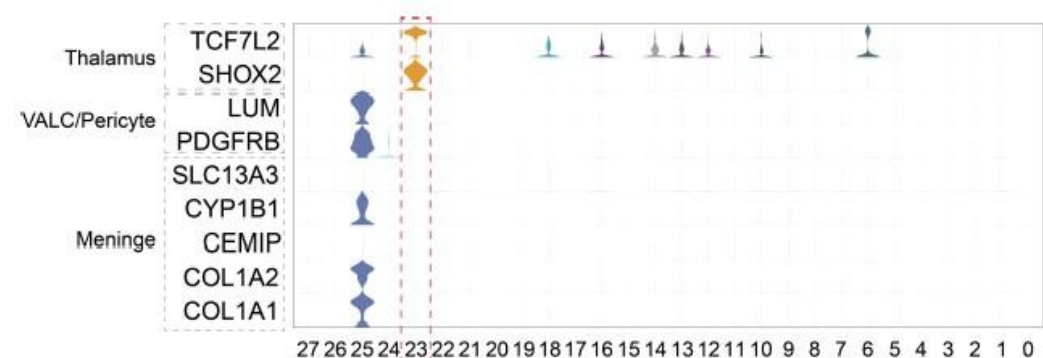
We appreciate for the reviewer's suggestion. We detected specific expression marker genes of cluster 20, cells in this cluster specifically expressed VIM, IGFBP5 and TAC. According to the cell annotation results from a published study, we relabeled cluster 20 as ventral LGE-derived interneuron precursors (Yu, Yuan et al. Nat Neurosci. 2021. doi:10.1038/s41593-021-00940-3. PMID: 34737447.). Cluster 20 cells have been deleted in the new Figure 3A.

1. Figure 1B UMAP - it is unexpected that meningeal cells would cluster topographically closer to the excitatory neuron cluster (one could even argue that the meningeal cell cluster is located within the excitatory neuron clusters) instead of next to or with the endothelial cell clusters. This is suspicious for a mis-annotated cell cluster. ZIC2 and ZIC3 were used as the principal marker genes for meningeal cells. However, these genes are not specific for meninges (PanglaoDB) and had not been identified as marker genes in a developmental sc-RNAseq dataset of the developing mouse meninges (DeSisto et al. 2020). Additional marker genes (COL1A1, COL1A2, CEMIP, CYP1B1, SLC13A3) may be helpful to delineate the identity of this cluster and provide more evidence for a meningeal origin.

We thank the reviewer for the constructive advice. The violin plot in Author response image 3 has checked additional marker genes, including COL1A1, COL1A2, CEMIP, and CYP1B2. Cluster 23 does not express these marker genes but specifically expresses thalamus marker genes SHOX2 (Rosin, Jessica M et al. Dev Biol. 2015. doi:10.1016/j.ydbio.2014.12.013. PMID: 25528224.) and TCF7L2 (Lipiec, Marcin Andrzej et al. Development. 2020. doi: 10.1242/dev.190181. PMID: 32675279). According to the gene expression results, we corrected the cell definition of cluster 23 to thalamic cells in the revised manuscript. Specifically, we added marker genes SHOX2 and CYP1B1 in the new Figure 1D violin plot and corrected the cell definition of cluster23 from meninges to thalamus cells in the revised manuscript and figures.

Author response image 3.

Violin plot of additional markers in cluster 23.



1. From Figure 1A, it appears that astrocytes (cluster 13) are present at E40 and E50 timepoints. This is inconsistent with literature and experimental data of the timing of the neuron-glia switch in primates and inconsistent with the claim within the text that, "Collectively, these results suggested that cortical neural progenitors undergo neurogenesis processes during the early stages of macaque embryonic cortical development, while gliogenic differentiation... occurs in later stages." The clarification of the percentage of astrocytes at each timepoint would clarify this point.

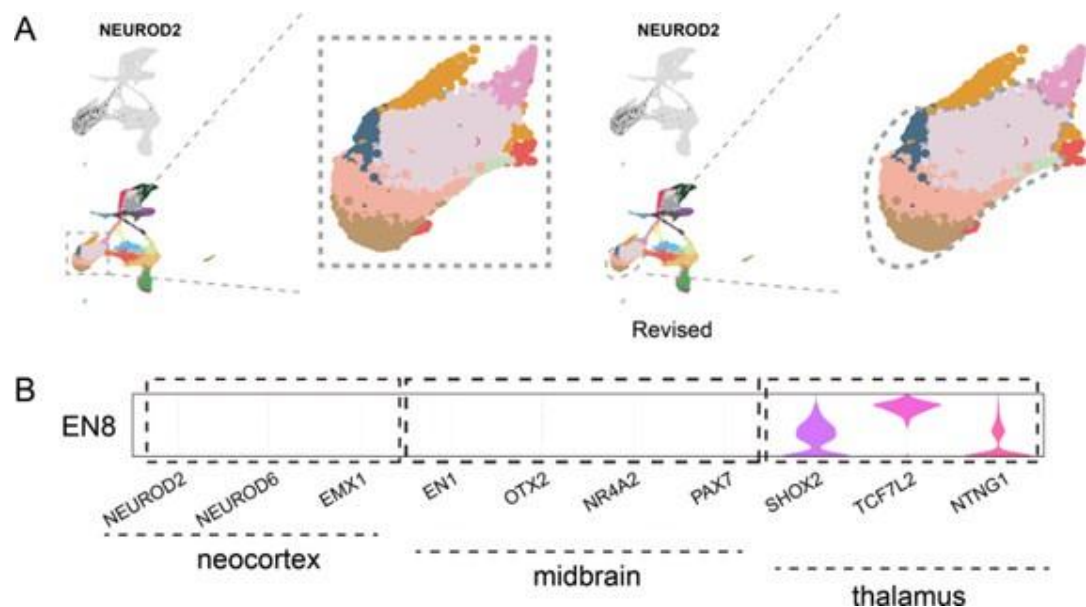
According to the suggestion, we have statistically analyzed the percentage of astrocytes (cluster 13) at each time point. The statistical results showed that the proportion of astrocytes was low to 0.1783% and 0.1046% at E40 and E50 time points, and increased significantly at E80 and E90, suggesting the onset of macaque gliogenesis might be around embryonic 80 days to 90 days. The result was consistent with published research on the timing of the neuron-glia transition in primates (Rash, Brian G et al. Proc Natl Acad Sci U S A. 2019. doi:10.1073/pnas.1822169116. PMID: 30894491). Besides, we thought that the cells in cluster 13 captured at E40 to E50 time points, with a total number of less than 200, maybe astrocyte precursor cells expressing the AQP4 gene (Yang, Lin, et al. Neuroscience bulletin. 2022. doi:10.1007/s12264-021-00759-9. PMID: 34374948).

1. A subcluster of ExN neurons was identified and determined to be of midbrain origin based on expression of TCF7L2. Did this subcluster express other known markers of the developing midbrain (OTX2, LMX1A, NR4A2, etc...)? Additionally, does this subcluster suggest that the limits of the dissection extended to the midbrain in samples E40 and E50?

We apologize for the previous inadequacy of the excitatory neuron cell annotation. In the description of the previous version of the manuscript, we misidentified the cells of the EN8 as midbrain cells. Following the reviewer's suggestion, we verified the expression of more tissue-specific marker genes of EN8. As the violin diagram in Author response image 4 shows, other developing midbrain markers OTX2, NR4A2, and PAX7 did not express in EN8, but thalamus marker genes SHOX2, TCF7L2, and NTNG1 were highly expressed in EN8. Besides, dorsal cortex excitatory neuron markers NEUROD2, NEUROD6, and EMX1 were not expressed in EN8, which suggests that EN8 might not belong to cortical cells. After carefully reviewing the data analysis process, we determined that EN8 was a small group of cells in cluster 23 mistakenly selected during excitatory neuron analysis, as shown in Figure R5(A), which was corrected after revision. In the revised manuscript, we have removed EN8 from the analysis of excitatory neurons. In the revised manuscript, we have deleted the previous EN8 subcluster and renumbered the left excitatory neuron subclusters in new Figure 2 and Figure S3.

Author response image 4.

(A). Modified diagram of clustering of excitatory neuron subclusters collected at all time points, visualized via UMAP related to Figure 2A. (B) Vlnplot of different marker genes in EN8.



1. "These data suggested that the cell fate determination by diverse neural progenitors occurs in the embryonic stages of macaque cortical development and is controlled by several key transcriptional regulators" The authors present a list of differentially expressed genes specific to the various radial glia clusters along pseudotime. Some of these radial glia DEGs are known and have been characterized by previous literature while other DEGs they have identified had not been previously shown to be associated with radial glia specification/maturation. However, this list of DEGs does not support the claim that cell fate determination is controlled by several key transcriptional regulators. What were the transcriptional regulators of radial glia specification identified in this study and how were they validated?

We agree with the reviewer and honestly admit that the description of this part in the previous manuscript is inaccurate. The description has been deleted in the revised manuscript.

1. "Comparing vRG to IPC trajectory between human, macaque, and mouse, we found this biological process of vRG-to-IPC is very conserved across species, but the vRG to oRG trajectory is divergent between species. The latter process is almost invisible in mice, but it is very similar in primates and macaque." Firstly, macaques are primates, and the text should be updated to reflect this. Secondly, from Figure 5C., it seems there were no outer radial glia detected at all within the vRG-oRG and vRG-IPC developmental trajectories. This would imply that oRGs are not "almost invisible" in mice, but rather do not exist. The authors need to clarify the presence or absence of identifiable outer radial glia in the integrated dataset and relate the relative abundance of these cells to their interpretation of the developmental trajectories for each species.

We apologize for the description inaccuracies in the manuscript and thank the reviewer for pointing out the expression errors. At your two suggestions, the description has been corrected in the revised manuscript as "Comparing vRG to IPC trajectory between human, macaque, and mouse, we found this biological process of vRG-to-IPC is very conserved across species. However, the vRG to oRG trajectory is divergent between species because the oRG population was not identified in the mouse dataset. The latter process is almost invisible in mice but similar in humans and macaques".

Although several published research has shown that oRG-like progenitor cells were present in the mouse embryonic neocortex(Wang, Xiaoqun et al. Nature neuroscience.2011. doi:10.1038/nn.2807; Vaid, Samir et al. Development. 2018, doi:10.1242/dev.169276. PMID: 30266827). However, oRG cells were barely detected in the scRNA-seq dataset of mice cortical development studies(Ruan, Xiangbin et al. Proc Natl Acad Sci U S A. 2021. doi:10.1073/pnas.2018866118. PMID: 33649223; Di Bella, Daniela J et al. Nature. 2021. doi:10.1038/s41586-021-03670-5. PMID: 34163074; Chen, Ao et al. Cell. 2022. doi:10.1016/j.cell.2022.04.003. PMID: 35512705). There were no oRG populations detected in the mouse embryonic cortical development dataset (GEO: GSE153164) used for integration analysis in our study.

1. *"Ventral radial glia cells generate excitatory neurons by direct and indirect neurogenesis"*
This should be corrected to dorsal radial glia cells as this paper is discussing radial glia of the dorsal pallium.

1. Editorially, gene names need to be italicized in the text, figures, and figure legends.

1. Figure 5B - a scale bar showing the scale of the relative expression denoted by the dark blue color would be beneficial.

1. Figure S7D is mislabeled in the figure legend.

Merged response to points 11 to 15: Thank you for kindly pointing out the errors in our manuscript. We have corrected the above four points in the revised version.

Reviewer #2 (Recommendations For The Authors):

Specific suggestions for authors:

In the abstract the authors state: "thicker upper-layer neurons". I think it's important to be clear in the language by stating either that the layers are thicker or the neurons are most dense.

Thanks for your good comments. The description of "thicker upper-layer neurons" was corrected to "the thicker supragranular layer" in the revised manuscript. The supragranular layer thickness in primates was much higher than in rodents, both in absolute thickness and in proportion to the thickness of the whole neocortex (Hutsler, Jeffrey J et al. Brain research. 2005. doi:10.1016/j.brainres.2005.06.015. PMID: 16018988). Here, we want to describe the supragranular layer of primates as significantly higher than that of rodents, both in absolute thickness and in proportion to the thickness of the whole neocortex.

The introduction needs additional clarification regarding the vRG vs oRG discussion. I was unclear what the main takeaway for readers should be. Similarly, the discussion of previous studies and the importance for comparing human and macaque could be clarified.

We appreciate the suggestion and apologize for the shortcomings of the introduction part. We have rewritten the section and added additional clarification in the revised introduction. In the revised manuscript, the contents of the introduction are as follows:

"The neocortex is the center for higher brain functions, such as perception and decision-making. Therefore, the dissection of its developmental processes can be informative of the mechanisms responsible for these functions. Several studies have advanced our

understanding of the neocortical development principles in different species, especially in mice. Generally, the dorsal neocortex can be anatomically divided into six layers of cells occupied by distinct neuronal cell types. The deep-layer neurons project to the thalamus (layer VI neurons) and subcortical areas (layer V neurons), while neurons occupying more superficial layers (upper-layer neurons) preferentially form intracortical projections¹. The generation of distinct excitatory neuron cell types follows a temporal pattern in which early-born neurons migrate to deep layers (i.e., layers V and VI), while the later-born neurons migrate and surpass early-born neurons to occupy the upper layers (layers II-IV)². In *Drosophila*, several transcription factors are sequentially explicitly expressed in neural stem cells to control the specification of daughter neuron fates, while very few such transcription factors have been identified in mammals thus far. Using single-cell RNA sequencing (scRNA-seq), Telley and colleagues found that daughter neurons exhibit the same transcriptional profiles of their respective progenitor radial glia, although these apparently heritable expression patterns fade as neurons mature³. However, the temporal expression profiles of neural stem cells and the contribution of these specific temporal expression patterns in determining neuronal fate have yet to be wholly clarified in humans and non-human primates. Over the years, non-human primates (NHP) have been widely used in neuroscience research as mesoscale models of the human brain. Therefore, exploring the similarities and differences between NHP and human cortical neurogenesis could provide valuable insight into unique features during human neocortex development.

In mammals, radial glial cells are found in the ventricular zone (VZ), where they undergo proliferation and differentiation. The neocortex of primates exhibits an extra neurogenesis zone known as the outer subventricular zone (OSVZ), which is not present in rodents. As a result of evolution, the diversity of higher mammal cortical radial glia populations increases. Although ventricular radial glia (vRG) is also found in humans and non-human primates, the vast majority of radial glia in these higher species occupy the outer subventricular zone (OSVZ) and are therefore termed outer radial glia (oRG). Outer radial glial (oRG) cells retain basal processes but lack apical junctions⁴ and divide in a process known as mitotic somal translocation, which differs from vRG⁵. vRG and oRG are both accompanied by the expression of stem cell markers such as PAX6 and exhibit extensive self-renewal and proliferative capacities⁶. However, despite functional similarities, they have distinct molecular phenotypes. Previous scRNA-seq analyses have identified several molecular markers, including HOPX for oRGs, CRYAB, and FBXO32 for vRGs⁷. Furthermore, oRGs are derived from vRGs, and vRGs exhibit obvious differences in numerous cell-extrinsic mechanisms, including activation of the FGF-MAPK cascade, SHH, PTEN/AKT, and PDGF pathways, and oxygen (O₂) levels. These pathways and factors involve three broad cellular processes: vRG maintenance, spindle orientation, and cell adhesion/extracellular matrix production⁸.

Some transcription factors have been shown to participate in vRG generation, such as INSM and TRNP1. Moreover, the cell-intrinsic patterns of transcriptional regulation responsible for generating oRGs have not been characterized.

ScRNA-seq is a powerful tool for investigating developmental trajectories, defining cellular heterogeneity, and identifying novel cell subgroups⁹. Several groups have sampled prenatal mouse neocortex tissue for scRNA-seq^{10,11}, as well as discrete, discontinuous prenatal developmental stages in human and non-human primates^{7,12,13,14}. The diversity and features of primate cortical progenitors have been explored^{4,6,7,15}. The temporally divergent regulatory mechanisms that govern cortical neuronal diversification at the early postmitotic stage have also been focused on¹⁶. Studies spanning the full embryonic neurogenic stage in the neocortex of humans and other primates are still lacking. Rhesus macaque and humans share multiple aspects of neurogenesis, and more importantly, the rhesus monkey and human brains share more similar gene expression patterns than the brains of mice and humans¹⁷⁻¹⁹. To establish a comprehensive, global picture of the

neurogenic processes in the rhesus macaque neocortex, which can be informative of neocortex evolution in humans, we sampled neocortical tissue at five developmental stages (E40, E50, E70, E80, and E90) in rhesus macaque embryos, spanning the full neurogenesis period. Through strict quality control, cell type annotation, and lineage trajectory inference, we identified two broad transcriptomic programs responsible for the differentiation of deep-layer and upper-layer neurons. We also defined the temporal expression patterns of neural stem cells, including oRGs, vRGs, and IPs, and identified novel transcription factors involved in oRG generation. These findings can substantially enhance our understanding of neocortical development and evolution in primates.”

Why is this study focused on the parietal lobe? This should be discussed in the introduction and interpretation of the data should be contextualized in the context of this cortical area.

In this study, samples were collected from the parietal lobe area mainly for the following reasons:

(1) To ensure that the cortical anatomical parts collected at each time point are consistent, we used the lateral cerebral sulcus as a marker to collect the parietal lobe tissue above the lateral sulcus for single-cell sequencing sample collection. Besides, the parietal region is also convenient for sampling the dorsal cortex.

(2) Previous studies have made the timeline of the macaque parietal lobe formation process during the prenatal development stage clear (Finlay, B L, and R B Darlington. Science. 1995. doi:10.1126/science.7777856. PMID: 7777856), which is also an essential reason for using the parietal lobe as the research object.

Figure 1:

Difficult to appreciate how single cell expression reflects the characterization of layers described in Figure 1A. A schematic for temporal development would be helpful. Also, how clusters correspond to discrete populations of excitatory neurons and progenitors would improve figure clarity. Perhaps enlarge and annotate the UMAPs on the bottom of Figure 1A.

We thank the reviewer for the suggestion and apologize for that Figure 1A does not convey the relationship between single-cell expression and neocortex layer formation. In the revised manuscript, time points information associated with the hierarchy is labeled to the diagram in Figure S1A. The UMAPs on the bottom of Figure 1A was enlarged in the revised manuscript as new Figure 1C.

Labels on top of clusters for 1A/1B would be helpful as it's difficult to see which colors the numbers correspond to on the actual UMAP.

Many thanks to the reviewer for carefully reading and helpful suggestions. We have adjusted the visualization of UMAP in the revised version. The numbers in the label bar of Figure 1B have been moved to the side of the dot so that the dot can be seen more clearly.

Microglia and meninges are also non-neural cells. This needs to be changed in the discussion of the results.

Thanks for the suggestion. We have fixed the manuscript as the reviewer suggested. The description in the revised manuscript has been fixed as follows: “According to the expression of the marker genes, we assigned clusters to cell type identities of neurocytes (including radial glia (RG), outer radial glia (oRG), intermediate progenitor cells (IPCs), ventral precursor

cells (VP), excitatory neurons (EN), inhibitory neurons (IN), oligodendrocyte progenitor cells (OPC), oligodendrocytes, astrocytes, ventral LGE-derived interneuron precursors and Cajal-Retzius cells, or non-neuronal cell types (including microglia, endothelial, meninge/VALC(vascular cell)/pericyte, and blood cells). Based on the expression of the marker gene, cluster 23 was identified as thalamic cells, which are small numbers of non-cortical cells captured in the sample collection at earlier time points. Each cell cluster was composed of multiple embryo samples, and the samples from similar stages generally harbored similar distributions of cell types.”.

It's important to define the onset of gliogenesis in the text and figure. What panels/ages show this?

We identified the onset of gliogenesis by statistically analyzing the percentage of astrocytes (cluster 13) at each time point and added the result in Figure S1. The statistical results showed that the proportion of astrocytes was deficient at E40 and E50 time points and increased significantly at E80 and E90, suggesting the onset of macaque gliogenesis might be around embryonic 80 days to 90 days. The result was consistent with published research on the timing of the neuron-glia transition in primates (Rash, Brian G et al. Proceedings of the National Academy of Sciences of the United States of America 201. doi:10.1073/pnas.1822169116. PMID: 30894491).

Figure 2:

Why are there so few neurons at E90? Is it capture bias, dissociation challenges (as postulated for certain neuronal subtypes in the discussion), or programmed cell death at this time point?

We thought it was because mature neurons at E90 with abundant axons and processes were hard to settle into micropores of the BD method for single cell capture. Due to the fixed size of the BD Rhapsody microwells, this sing-cell capture method might be less efficient in capturing mature excitatory neurons but has a good capture effect on newborn neurons at each sampling time point. In conclusion, based on the BD cell capture method feature, the immature neurons at each point are more easily captured than mature neurons in our study, so the generation of excitatory neurons at different developmental time points can be well observed, as shown in Figure 2, which aligns with our research purpose.

The authors state: "We then characterized temporal changes in the composition of each EN subcluster. While the EN 5 and EN 11 (deep-layer neurons) subclusters emerged at E40 and E50 and disappeared in later stages, EN subclusters 1, 2, 3, and 4 gradually increased in population size from E50 to E80 (Figure 2D)." What about EN7? It's labeled as an upper layer neuron that is proportionally highest at E40. Could this be an interesting, novel finding? Does this indicate something unique about macaque corticogenesis? The authors don't describe/discuss this cell type at all.

We apologize for the manuscript's lack of detailed descriptions of EN results. In our study, EN7 is identified as CUX1-positive, PBX3-positive, and ZFHX3-positive excitatory neuron subcluster. The results of Fig. 2B show that EN7 was mainly captured from the early time points (E40/E50) samples. Above description was added in the revised manuscript.

The Pbx/Zfhx3-positive excitatory neuron subtype reported in Moreau et al. study on mouse neocortex development progress (Moreau, Matthieu X et al. Development. 2021. doi:10.1242/dev.197962. PMID: 34170322) . Our study verified that the Pbx3/Zfhx3-positive cortical excitatory neurons also exist in the early stage of prenatal macaque cortex development.

Is there any unique gene expression in identified subtypes that are surprising? Did the comparison against human data, in later figures, inform any unique features of gene expression?

Based on the excitatory neuron subclusters analysis result in our study, we found no astonishing results in excitatory neuron subclusters. In subsequent integrated cross-species analyses, macaque excitatory neurons showed similar transcriptional characteristics to human excitatory neurons. In general, excitatory neurons tend to have a greater diversity in the cortex of animals that are more advanced in evolution (Ma, Shaojie et al. Science. 2022. doi:10.1126/science.abo7257. PMID: 36007006; Wei, Jia-Ru et al. Nat Commun. 2022. doi:10.1038/s41467-022-34590-1. PMID: 36371428; Galakhova, A A et al. Trends Cogn Sci. 2022. doi:10.1016/j.tics.2022.08.012. PMID: 36117080; Berg, Jim et al. Nature. 2021. doi:10.1038/s41586-021-03813-8. PMID: 34616067). Since only single-cell transcriptome data was analyzed in this study, we did not find any unique features of the prenatal developing macaque cortex excitatory neurons in the comparison against the human dataset due to the limitation of information dimension.

Figure 3:

The identification of terminal oRG differentiation genes is interesting. The confirmation of known gene expression as well as novel markers that indicate different states/stages of oRG cells is a valuable resource. As the identification of described ion channel expression is a novel finding, it should be explored more and would be strengthened by validation in tissue samples and, if possible, functional assays.

E is the most novel part of this figure, but it's very hard to read. I think increasing the focus of this figure onto this finding and parsing these results more would be informative.

Thanks for the positive comments. We apologize for the lack of clarity and conciseness in figure visualizations. We hypothesized vRG to oRG cell trajectories into three phases: onset, commitment, and terminal. The leading information conveyed by Figure 3E was the dynamic gene expression along the developmental trajectory from vRG to oRG. Specific genes were selected and shown in the schema diagram of new Figure 3.

We verified terminal oRG differentiation genes in the recently published macaque telencephalic development dataset (Micali N, Ma S, Li M, et al. Science. doi:10.1126/science.adf3786. PMID: 37824652) (GEO accession: GSE226451). The results of Author response image 2 show that the gene expression showed states/stages. Most of the oRG terminal differentiation markers genes identified in our study were also expressed in the oRG cells of the GSE226451 dataset. In particular, the two datasets were consistent in the expression of ion channel genes ATP1A2, ATP1A2, and SCN4B.

I'm curious about the granularity of the oRG_C12 terminal cluster. Are there ways to subdivide the different cells that seem to be glial-committed vs actively dividing vs neurogenically committed to IPCs? In the text, the authors referred to different oRG populations, but they are annotated as the same cluster and cell type. The authors should clarify this.

According to the reviewer's suggestion, we subdivide the oRG_C12 into eight subclusters. Based on the marker gene in Author response image 5C, subclusters 1,2 and 4 might be glial-committed with AQP4/S100B positive expression; subclusters 3,6,7 might be neurogenically committed to IPCs with NEUROD6 positive expression; subclusters 0,3,5,6,7 might be actively dividing state with MKI67/TOP2A positive expression.

Author response image 5.

Subdivide analysis of oRG_C12. (A) and (B) Subdividing of e oRG_C12 visualized via UMAP. Cells are colored according to subcluster timepoint (A) and subcluster identities (B). (C) Violin plot of molecular markers for the subclusters.

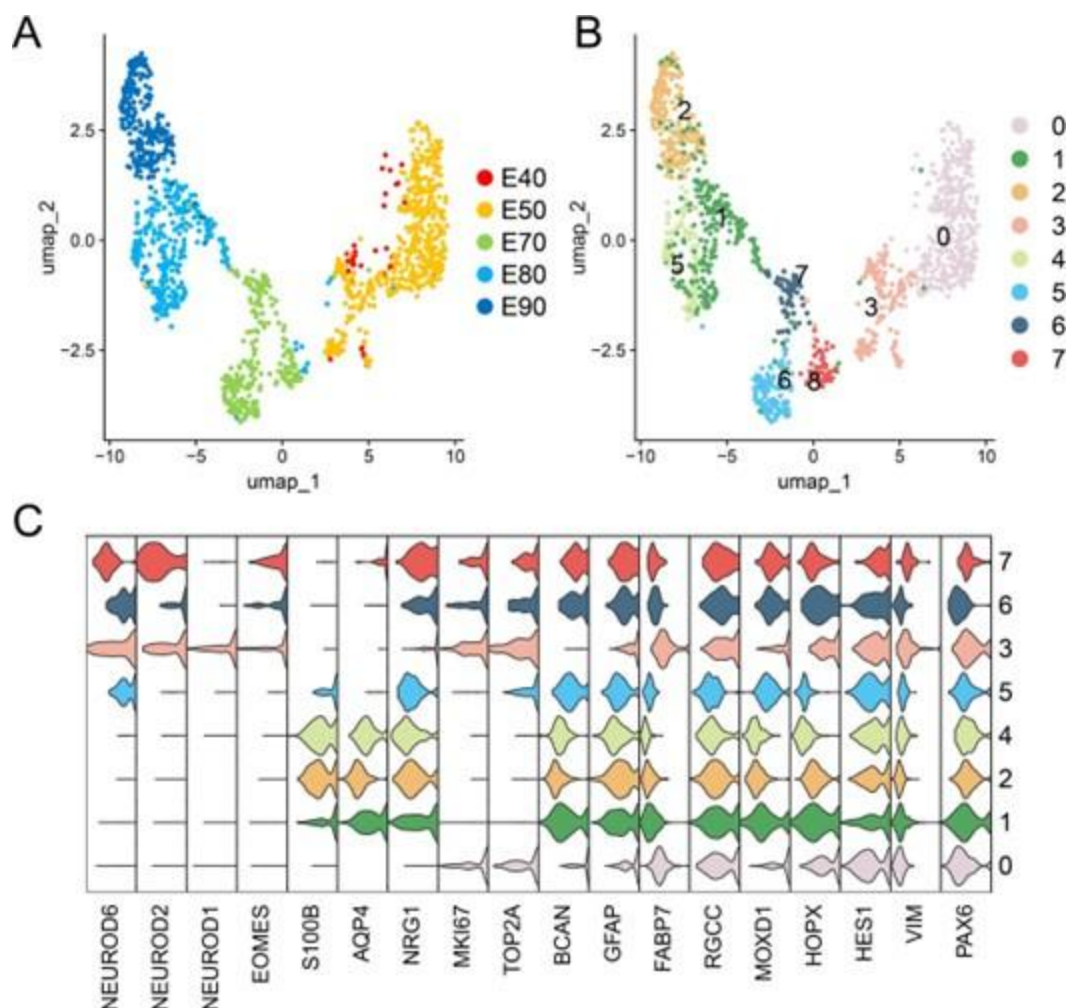


Figure 4:

Annotating/labeling the various EN clusters (even as deep/upper) would help improve the clarity of this and other figures. It's clear what each progenitor subtype is but it's hard to read the transitions. Why are all the EN groups in pink/red? It makes the data challenging to interpret.

In Figure4A, we use different yellow/orange colors for deep-layer excitatory neuron subclusters (EN5 and EN10), and different red/pink colors for upper-layer excitatory neuron subclusters (EN1, EN2, EN3, EN4, EN6, EN7, EN8 and EN9). We add the above information in the legend of Figure 4 in the revised manuscript.

E50 seems to be unique - what's EN11?

Based on the molecular markers for EN subclusters in Author response image 2, we recognized EN11 as a deep-layer excitatory neuron subcluster expressing BCL11B and FEZF2.

As explained in the above reply, the microplate of BD has a good effect on capturing newborn neurons at each time point. The EN11 was mainly a newborn excitatory neuron at the E50 timepoint, which makes the subcluster seem unique.

Author response image 6.

Vlnplot of different markers in EN8.



Figure 4E - the specificity of gene expression for deep vs upper layer markers seems to be over stated given the visualized gene expression pattern (ex FEZF2). Could the right hand panels be increased to better appreciate the data and confirm the specificity, as described.

In our study, we used slingshot method to infer cell lineages and pseudotimes, which have been used to identifying biological signal for different branching trajectories in many scRNA-seq studies. We apologize for the lack of visualization clarity in the figure 4E. Due to the size limitation of the uploaded file, the file was compressed, resulting in a decrease in the clarity of the image. Below, we provided figure 4E with a higher definition and increased several genes' slingshot branching tree results according to the reviewer's suggestion.

Figure 5:

There are some grammatical typos at the bottom of page 8. In this section, it also feels like there is a missing logical step between expansion of progenitors through elongated developmental windows that impact long-term expansion of the upper cortical layers.

We apologize for the grammatical typos and have corrected them in the revised manuscript. We understand the reviewer's concern. Primates have much longer gestation than rodents, and previous study evidence had shown that extending neurogenesis by transplanting mouse embryos to a rat mother increases explicitly the number of upper-layer cortical neurons, with concomitant abundant neurogenic progenitors in the subventricular zone (Stepien, Barbara K et al. Curr Biol. 2020. doi:10.1016/j.cub.2020.08.046. PMID: 32888487). We thought this mechanism could also explain primates' much more expanded abundance of upper-layer neurons.

I'm curious about the IPCs that arise from the oRGs. Lineage trajectory shows vRG decision to oRG or IPC, but oRGs also differentiate into IPCs. Could the authors conjecture why they are not in this dataset or are indistinguishable from vRG-derived IPCs.

Several published experiments have proved that oRG can generate IPC in human and macaque developing neocortex. (Hansen, David V et al. Nature. 2010. doi:10.1038/nature08845. PMID: 20154730; Betizeau, Marion et al. Neuron. 2013. doi:10.1016/j.neuron.2013.09.032. PMID: 24139044). Clearly identifying the difference between IPC generated from vRG and oRG at the transcriptional level in our single-cell transcriptome dataset is difficult. We hypothesized that the IPCs produced by both pathways have highly similar transcriptional features. Due to the limit of the scRNA data analysis algorithm used in

this study, we didn't distinguish the two kinds of IPC, which could not be in terms of pseudo-time trajectory reconstruction and transcriptional data.

Figure 6 :

How are the types 1-5 in 6A defined? Were they defined in one species and then applied across the others?

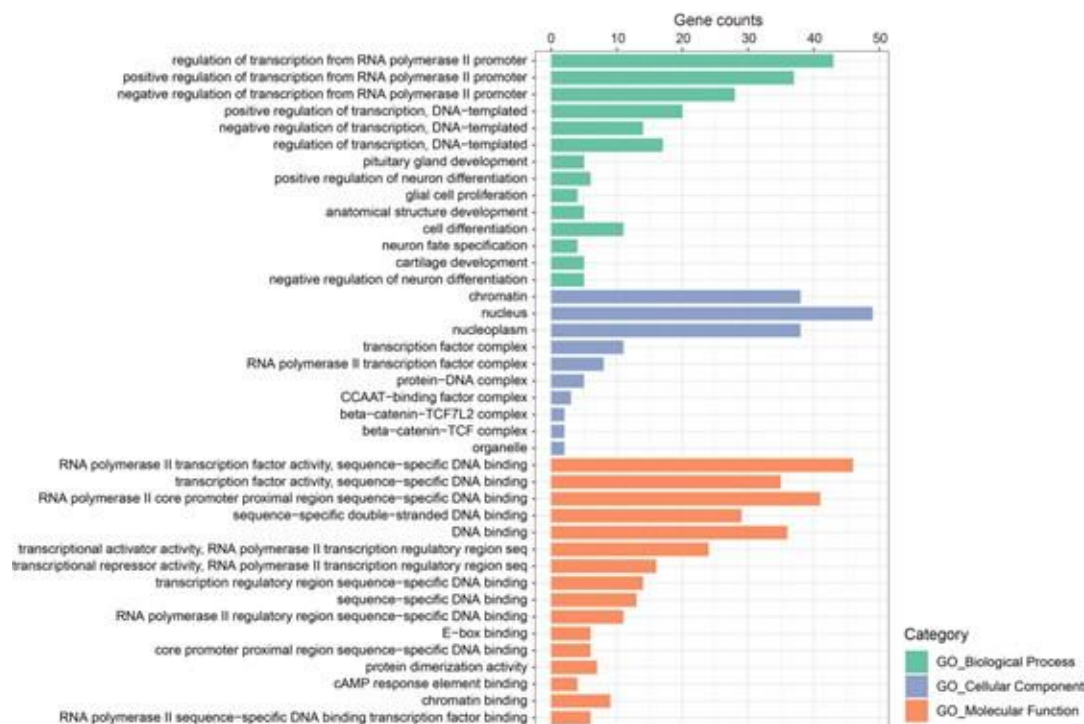
We applied the same analysis to each species. We first picked up vRG cells in each species dataset and screened the differentially expressed genes (DEGs) between adjacent development time points using the “FindMarkers” function (with min. pct = 0.25, logfc. threshold = 0.25). After separate normalization of the DEG expression matrix from different species datasets, we use the “standardise” function from the Mfuzz package to standardize the data. The DEGs of vRG in each species were grouped into five clusters using the Mfuzz package in R with fuzzy c- means algorithm.

The temporal dynamics in the highlighted section in B have interesting, consistent patterns of gene expression of the genes described, but what about the genes below that appear less consistent temporally? What processes do not appear to be conserved, given those gene expression differences?

Many thanks for the constructive comments. The genes in Figure 6B below are temporal dynamics non-conserved transcription factors among the three species vRG. We performed a functional enrichment analysis on the temporal dynamics of non-conserved transcription factors with the PANTHER (Protein ANalysis THrough Evolutionary Relationships) Classification System(<https://www.pantherdb.org/>), and the analysis results are shown in Author response image 7. The gene ontology (GO) analysis results show that unconserved transcription factors were related to different biological processes, cellular components, and molecular functions. However, subsequent experiments are still needed to verify specific genes.

Author response image 7.

Gene Ontology (GO) analysis of unconserved temporal patterns transcription factors among mouse, macaque and human vRG cells.



The identification of distinct regulation of gene networks, despite conservation of transcription factors in discrete cell types, is interesting. What does the comparison between humans and macaques indicate about regulatory differences evolutionarily?

We appreciate the reviewer for the comments. We performed the TFs regulation network analysis of human vRG with pyscenic workflow. The top transcription factors of every time point in human vRG were calculated, and we used the top 10 TFs and their top 5 target genes to perform interaction analysis and generate the regulation network of human vRG in revised figure 6. In comparison of the pyscenic results of mouse, macaque and human vRG, it was obvious that the regulatory networks were not evolutionarily conservative. Compared with macaque, the regulatory network of transcription factors and target genes in humans is more complex. Some conserved regulatory relationships present in more than one species are identified, such as HMGN3, EMX2, SOX2, and HMGA2 network at an early stage when deep layer generation and SOX10, ZNF672, ZNF672 network at a late stage when upper-layer generation.

Reviewer #3 (Recommendations For The Authors):

The data should be compared to a similar brain region in human and mouse, if available. (See data from PMID: PMC8494648).

We appreciate the reviewer's suggestions. In Figure 6, the species-integration analysis, the mouse data were from the perspective of the somatosensory cortex, macaque data were mainly from the parietal lobe in this study, and human data including the frontal lobe (FL), parietal lobe (PL), occipital lobe (OL), and temporal lobe (TL). PMC8494648 offered high-quality data covering the period of gestation week 14 to gestation week 25. However, our study's development stage of rhesus monkeys is E40-E90 days, corresponding to pcw8-pcw21 in humans. The quality of data from PMC8494648 is particularly good. However, the developmental processes covered by PMC8494648 don't perfectly match the development time of the macaque cortex that we focused on in this study. Therefore, it is challenging to

integrate the dataset (PMCID: PMC8494648) into the data analysis part. However, we have cited the results of this precious research (PMCID: PMC8494648) in the discussion part of the revised manuscript.

A deeper assessment of these data in the context of existing studies would help distinguish the work and enable others to appreciate the significance of the work.

We appreciate the reviewer's constructive suggestions. The human regulation analysis with pyscenic workflow was added into new figure 6 for the comparison of different species vRG regulatory network. Analysis of the regulatory activity of human, macaque and mouse prenatal neocortical neurogenesis indicated that despite commonalities in the roles of classical developmental TFs such as GATA1, SOX2, HMGN3, TCF7L1, ZFX, EMX2, SOX10, NEUROG1, NEUROD1 and POU3F1. The top 10 TFs of the human, macaque, and mouse vRG each time point and their top 5 target genes identified by pySCENIC as an input to construct the transcriptional regulation network (Figure 6 D, F and H). Some conserved regulatory TFs present in more than one species are identified, such as HMGN3, EMX2, SOX2, and HMGA2 at an early stage when deep- lager generation and SOX10, ZNF672, and ZNF672 at a late stage when upper-lay generation.

Besides, we performed some comparative analysis with our macaque dataset and the newly published macaque telencephalon development dataset. The results were only used to provide additional information to reviewers and were not included in the revised manuscript.

To verify the reliability of our cell annotation results, we compared the similarity of cell-type association between our study and recently published research(Micali N, Ma S, Li M, et al. Science. doi:10.1126/science.adf3786.PMID: 37824652), using the scmap package to project major cell types in our macaque development scRNA-seq dataset to GSE226451. The river plot in Author response image 1 illustrates the broadly similar relationships of cell type classification between the two datasets. Otherwise, we used more marker genes for cell annotation to improve the results of cell type definition in new Figure 1D. Besides, the description of distinct excitatory neuronal types has been improved in the new Figure 2.

Furthermore, we verified terminal oRG differentiation genes in the recently published macaque telencephalic development dataset(Micali N, Ma S, Li M, et al. Science. doi:10.1126/science.adf3786.PMID: 37824652) (GEO accession: GSE226451). The results of Authro response image 2 show that the gene expression showed states/stages. Most of the oRG terminal differentiation markers genes identified in our study were also expressed in the oRG cells of the GSE226451 dataset. In particular, the two datasets were consistent in the expression of ion channel genes ATP1A2, ATP1A2, and SCN4B.